

# Self Organized Criticality

*Kim Sneppen, September 2025*



**Stuff that either move, or stay exactly put**

# First a Tutorial in Random Walks & Branching processes

$$x = \sum_{i=1}^{t/\tau} \eta(i) \quad \eta(i) \text{ takes values } \pm\ell \text{ randomly}$$

$$\langle x^2 \rangle = \left\langle \sum_{i=1}^{t/\tau} \eta(i) \sum_{j=1}^{t/\tau} \eta(j) \right\rangle$$

$$= \sum_{i,j=1}^{t/\tau} \langle \eta(i) \eta(j) \rangle$$

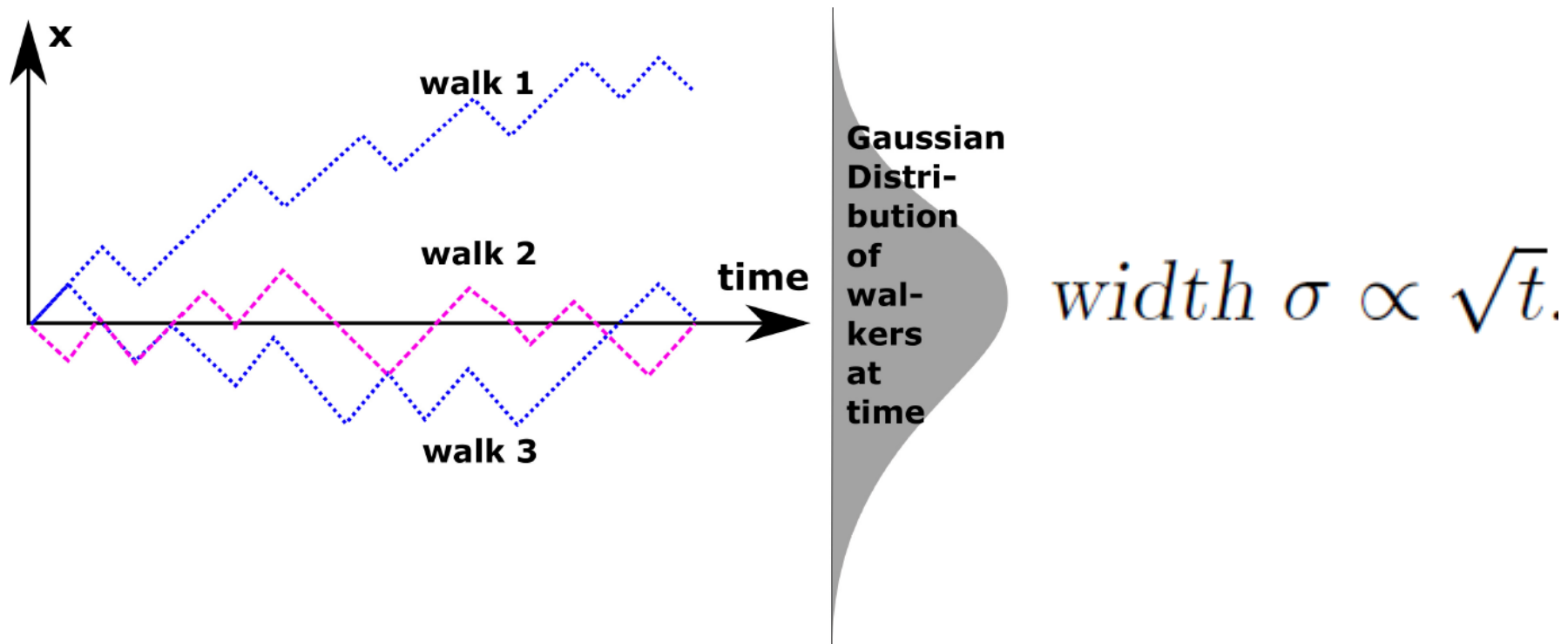
$$= \sum_{i=1}^{t/\tau} \langle \eta(i) \eta(i) \rangle$$

$$= \sum_{i=1}^{t/\tau} \ell^2 = (t/\tau) \ell^2 = (\ell^2/\tau) t = 2 \cdot D \cdot t$$

Diffusion constant  
Units [m\*m/sec].



# Random walks tend to stay around start point



What is density of crossings?:.....

## Relation to diffusion:

$$\rho(x, t + 1) - \rho(x, t) = \frac{1}{2}\rho(x - 1, t) - \frac{1}{2}\rho(x, t) - \frac{1}{2}\rho(x, t) + \frac{1}{2}\rho(x + 1, t)$$

$$\frac{\partial \rho}{\partial t} = D \frac{\partial^2 \rho}{\partial x^2}$$

$$D = 1/2$$

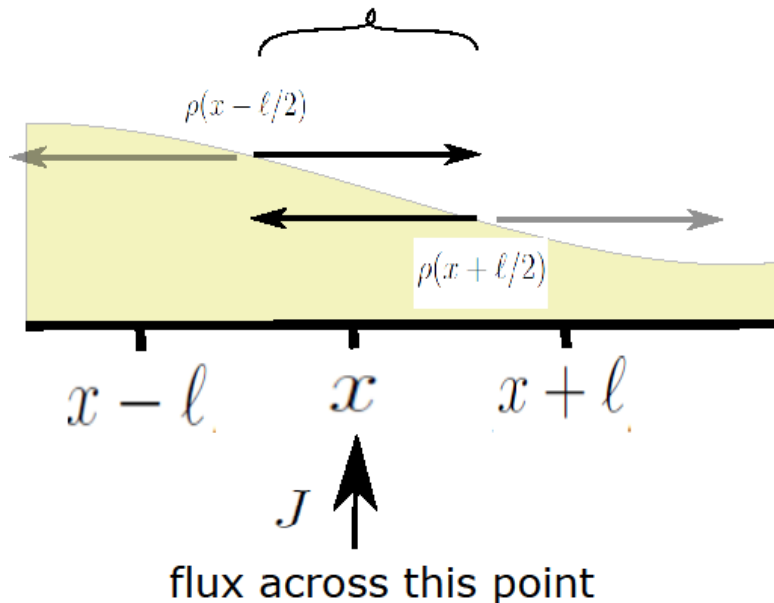
# Formal derivation of diffusion equation:

$$\begin{aligned}
 J &= \frac{1}{2} \left( \rho(x - \ell/2) \frac{\ell}{\tau} - \rho(x + \ell/2) \frac{\ell}{\tau} \right) \\
 &= \frac{\ell}{2\tau} \left( \left( \rho(x) - \frac{\ell}{2} \cdot \frac{d\rho(x)}{dx} \right) - \left( \rho(x) + \frac{\ell}{2} \frac{d\rho(x)}{dx} \right) \right) \\
 &= -D \cdot \frac{d\rho(x)}{dx}
 \end{aligned}$$

$$\frac{d\rho}{dt} = -\frac{dJ}{dx} = \frac{d}{dx} \left( D \frac{d\rho}{dx} \right)$$

$$\rho(x, t) = \frac{1}{\sqrt{4\pi t D}} \exp\left(-\frac{x^2}{4Dt}\right)$$

Move distance  $\ell$  during timestep  $\tau$



Where Diffusion constant:

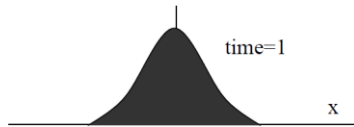
$$D = \ell^2 / (2\tau)$$

# Diffusion equation, solution:

$$\frac{d\rho}{dt} = \frac{d}{dx} \left( D \frac{d\rho}{dx} \right)$$

**Solution:**  $\rho(x, t) = \frac{1}{\sqrt{4\pi t D}} \cdot \exp\left(-\frac{x^2}{4Dt}\right)$

$= F \cdot E$  with:  $F \propto t^{-1/2}$  and  $E = \exp(-x^2/4Dt)$

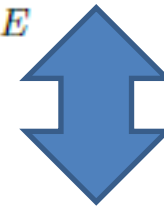


**Proff:**

$$d\rho/dx = -F \cdot \frac{x}{2Dt} \cdot E$$

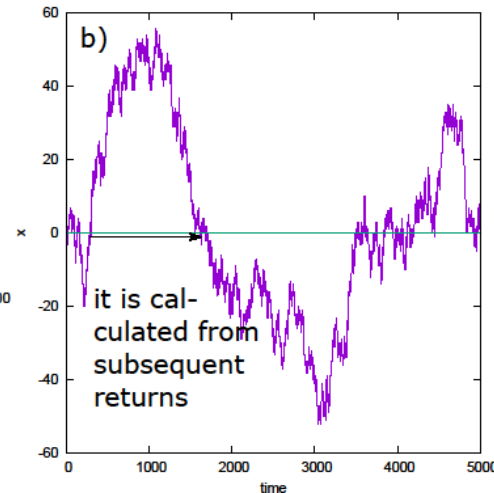
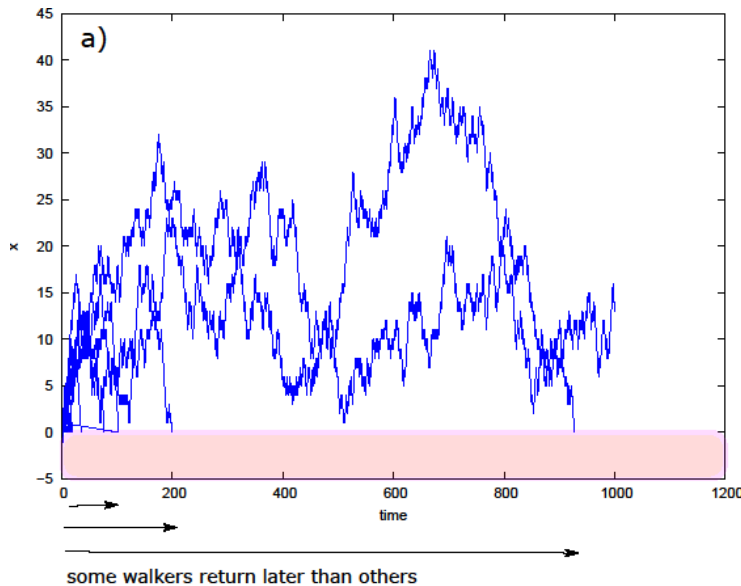
$$d^2\rho/dx^2 = -F \cdot \frac{1}{2Dt} \cdot E + F \cdot \left(\frac{x}{2Dt}\right)^2 \cdot E$$

$$d\rho/dt = -\frac{1}{2t} \cdot F \cdot G + F \cdot \frac{x^2}{4Dt^2} \cdot E$$



A factor D different

# First passage for random walks:



Simple derivation: **Assume power law** for first return:

$$t = \sqrt{t} \int_0^t t' P_{first}(t') dt'$$

Total  
time

Number  
of crossings  
In panel b)

Average time  
between crossings

$$P_{first}(t) \propto t^{-\tau} \longrightarrow t = t^{1/2+2-\tau}, \text{ or } P_{first}(t) = t^{-3/2}$$

# Analog to second fractal equation:

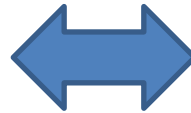
$$L^{D_{num}} \cdot \int_1^{L^D} s \cdot n(s) ds = L^{D_{tot}} \quad \longleftrightarrow \quad t = \sqrt{t} \int_0^t t' P_{first}(t') dt'$$

$$L^{D_{num}} \cdot [s^{2-\tau}]_1^{L^D} = L^{D_{tot}}$$

$$L^{D_{num} + (2-\tau)D} = L^{D_{tot}}$$

$$D_{num} + (2 - \tau) \cdot D = D_{tot}$$

$$\tau = 2 - \frac{D_{tot} - D_{num}}{D},$$



$$D_{tot}=1, D_{num}=0.5, D=1$$

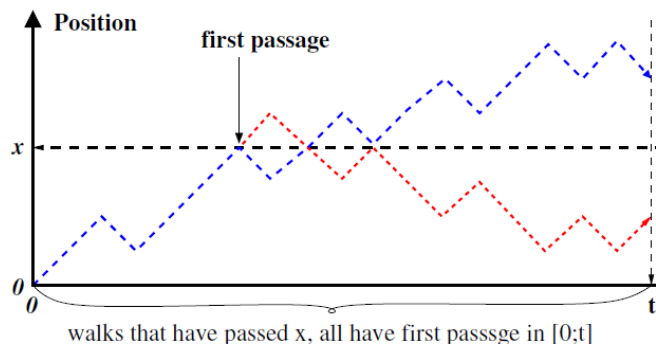


# Formal derivation

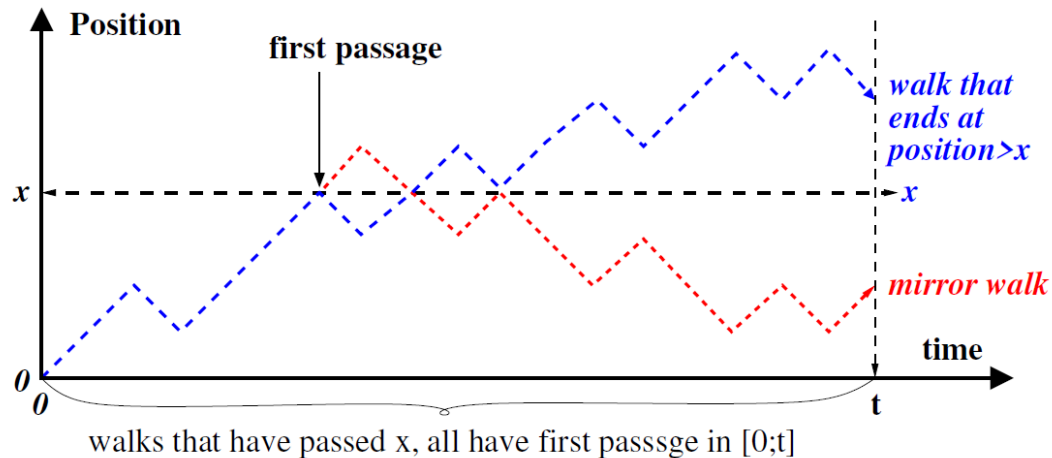
- Consider first passage times to  $x$ :
- And its cumulative distribution:

$$First_x(t)$$

$$\mathcal{P}_x(t) = \int_0^t First_x(t') dt'$$



- This cumulative is equal to the probability that the walker was at positions  $>x$  before  $t$
- Conjecture: For each walk that ends in  $y>x$  there is 2 walks that was larger than  $x$  before  $t$



For each walk that ends at position  $> x$  there is 2 walks that passed position  $x$  in interval  $[0;t]$

The 2 walks that was larger than  $x$  before  $t$  is:  
The **walk** and its **mirror walk** illustrated above

$$\mathcal{P}_x(t) = P(\max \text{ excursion in } [0;t] > x) = 2 \cdot P(\text{end position is } > x)$$

=Integrated  $P_{\text{first}} \rightarrow$   
Differentiate to get  $P_{\text{first}}$

$$= 2 \frac{1}{\sqrt{4\pi Dt}} \int_x^\infty dy \cdot e^{-y^2/4Dt}$$

$$\rho(x,t) = \frac{1}{\sqrt{4\pi tD}} \cdot \exp\left(-\frac{x^2}{4Dt}\right)$$

# The time to stay in a perfect casino:

$$\begin{aligned} First_x(t) &= \frac{d}{dt} \mathcal{P}_x(t) \\ &= 2 \frac{1}{\sqrt{4\pi D}} \frac{d}{dt} \left( \frac{1}{\sqrt{t}} \int_x^\infty dy e^{-y^2/4Dt} \right) \end{aligned}$$

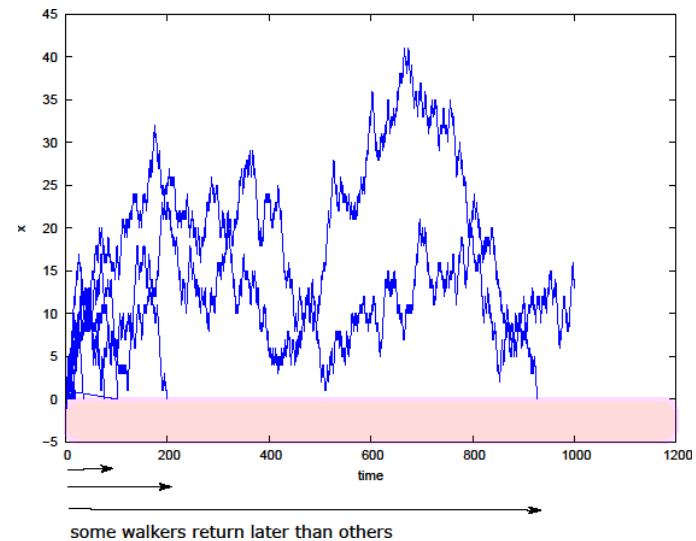
$$v = 2Dtx^2/y^2$$


$$dv = -\frac{4Dtx^2}{y^3} \cdot dy \Rightarrow \frac{dy}{\sqrt{t}} = -\frac{y^3 \cdot dv}{4Dx^2 t^{3/2}} = -\frac{1}{\sqrt{2Dx^2}} \cdot \frac{dv}{v^{3/2}} \propto -\frac{dv}{v^{3/2}}$$

$$First_x(t) \propto 2 \frac{d}{dt} \left( \int_0^{2Dt} \frac{dv}{v^{3/2}} \cdot e^{-x^2/2v} \right) \propto \frac{1}{t^{3/2}} \cdot \exp(-x^2/4Dt)$$

$$First_{x \sim 0}(t) \propto \frac{1}{t^{3/2}} \quad \text{for } x \ll \sqrt{4Dt}$$

# Consequence:



- Average time is infinite, ....why?
- Chance to cross zero declines with time,.... What is it that changes as time increases?
- What would you guess happens to your distribution  $1/t^{1.5}$  distribution if you win with slightly less probability than you lose? (Roulette have 18/37 to win and 19/37 to lose)
- Analog to critical percolation of Bethe lattice?

# Critical Branching:

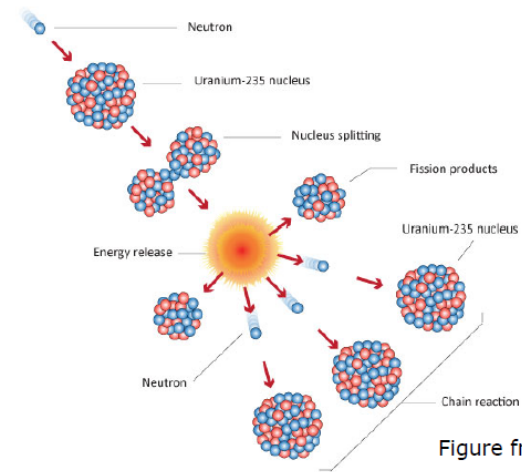


Figure from [passmyexams.co.uk/GCSE/physics](http://passmyexams.co.uk/GCSE/physics)

Super-Critical chain reaction: each reaction give rise to  $> 1$  new reaction

Critical chain reaction: each reaction give rise to one new reaction

Sub-Critical chain reaction: each reaction give rise to  $< 1$  new reaction

In critical branching each branch have 50% chance of giving 2 new branches, and 50% chance of dying out.

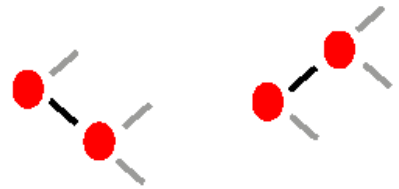
...

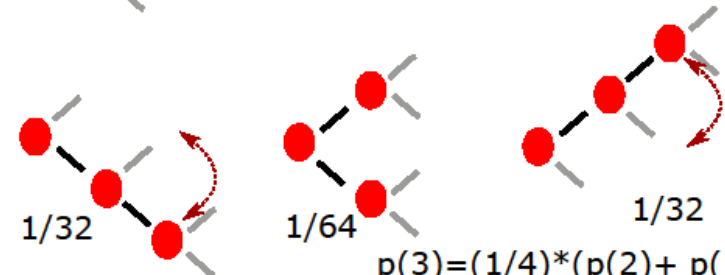
Number of live branches perform a random walk

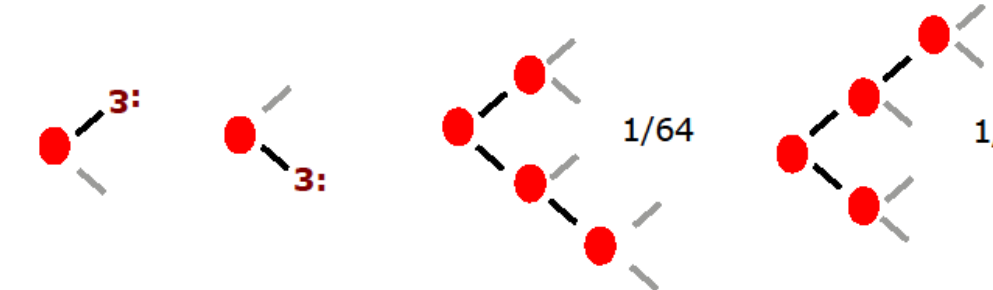
When counted in terms of the number of branch points.

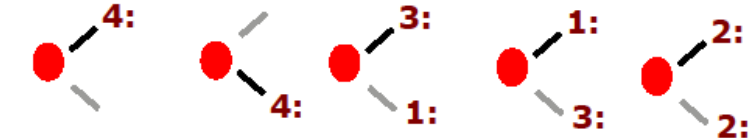
If you start with filled site:

1:   $p(1)=1/4$

2:   $p(2)=2/2^4=1/8$

3:   $p(3)=5/64$   
 $p(3)=(1/4)*(p(2)+ p(1)*p(1)+p(2))=5/64$

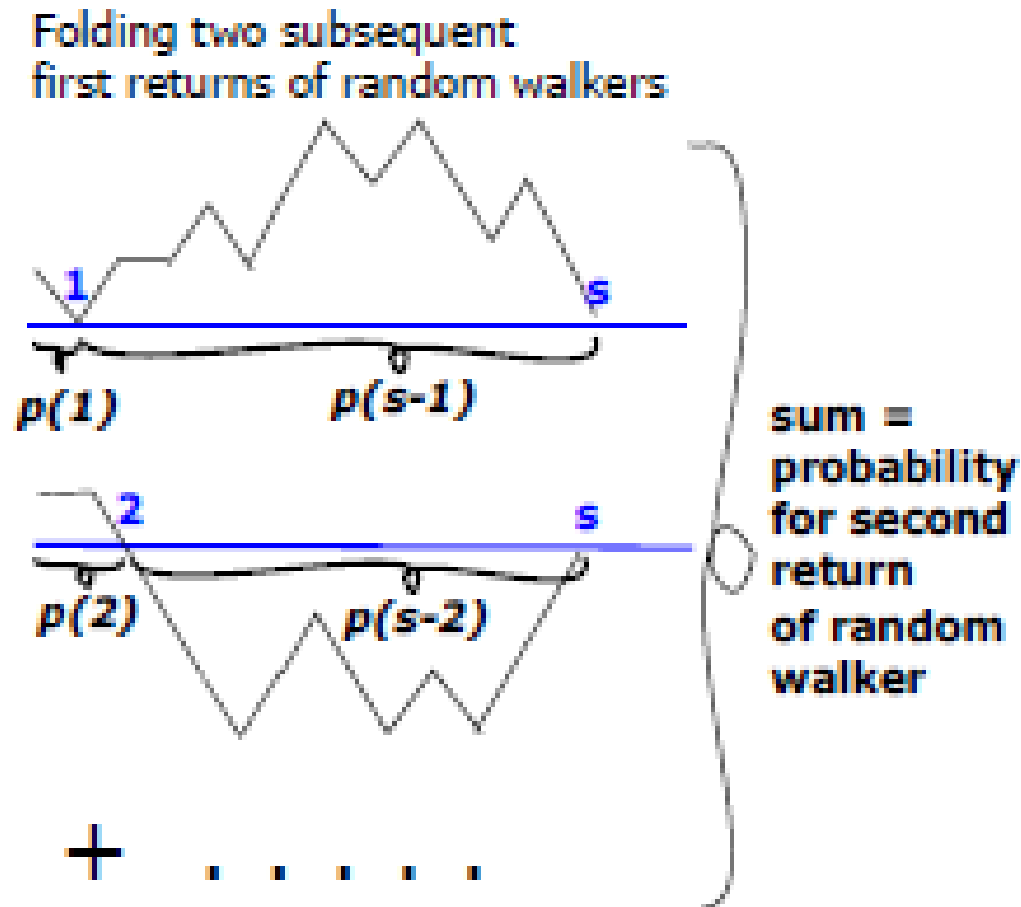
4:   $p(4)=7/256$

5:   $p(5)=(1/4)*(p(4)+p(3)*p(1)+p(2)*p(2)+p(1)*p(3)+p(4))$

$$p(s) = \frac{1}{4} \sum_{k=0}^{s-1} p(s-k-1) p(k)$$

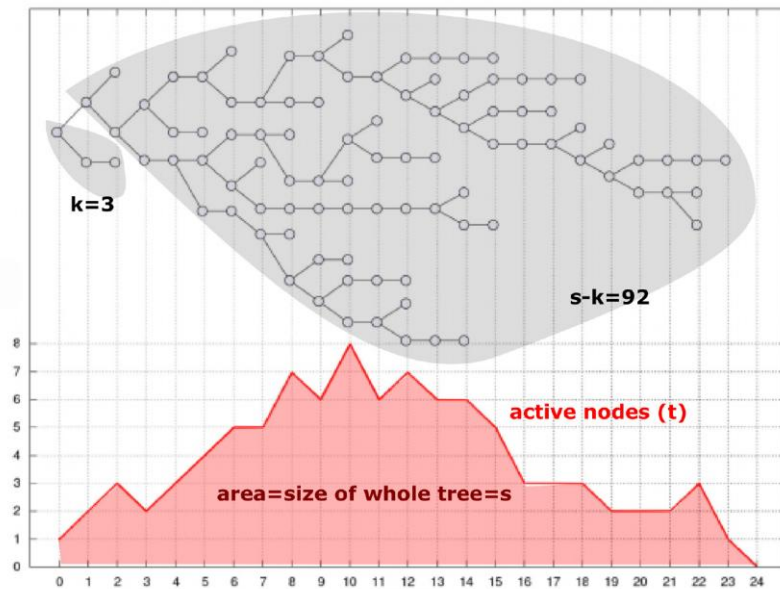
where we define  $p(0)=1$

# Random walker give nearly same recursion:



$$p_{second}(s) = \sum_{u=1}^{s-1} p(u)p(s-u)$$

# Critical branching-Random walk:



→ *Criticality (50%-50% decisions)* →  
 $P(s)$  of the whole cluster should be  
 Equal to  $p(s)$  of each of the below branches,  
 Approximately (see previous page):

$$p(s) \propto \sum_{k=1}^{s-1} p(k) \cdot p(s-k)$$

...also give second return of  
 random walker in terms of two first returns,  
 assuming that they are similar.

→  $p(s) \propto 1/s^{3/2}$

$$\begin{aligned}
 G(x) &= \sum_{i=1}^{\infty} p_i x^i \quad \rightarrow \quad \sum p_i p_{k-i} = 2p_k \quad \rightarrow \quad G^2(x) = \sum_{i=1}^{\infty} p_i x^i \sum_{j=1}^{\infty} p_j x^j \quad \rightarrow \quad G^2(x) = 2 \cdot (G(x) - p_1 \cdot x) \Rightarrow G(x) = 1 - \sqrt{1-x} \\
 & \quad \quad \quad G(1) = \sum_i p_i = 1 \\
 & \quad \quad \quad = \sum_{k=2}^{\infty} \sum_{i=1}^{k-1} p_i p_{k-i} x^k \\
 & \quad \quad \quad = \sum_{k=2}^{\infty} 2p_k x^k,
 \end{aligned}$$

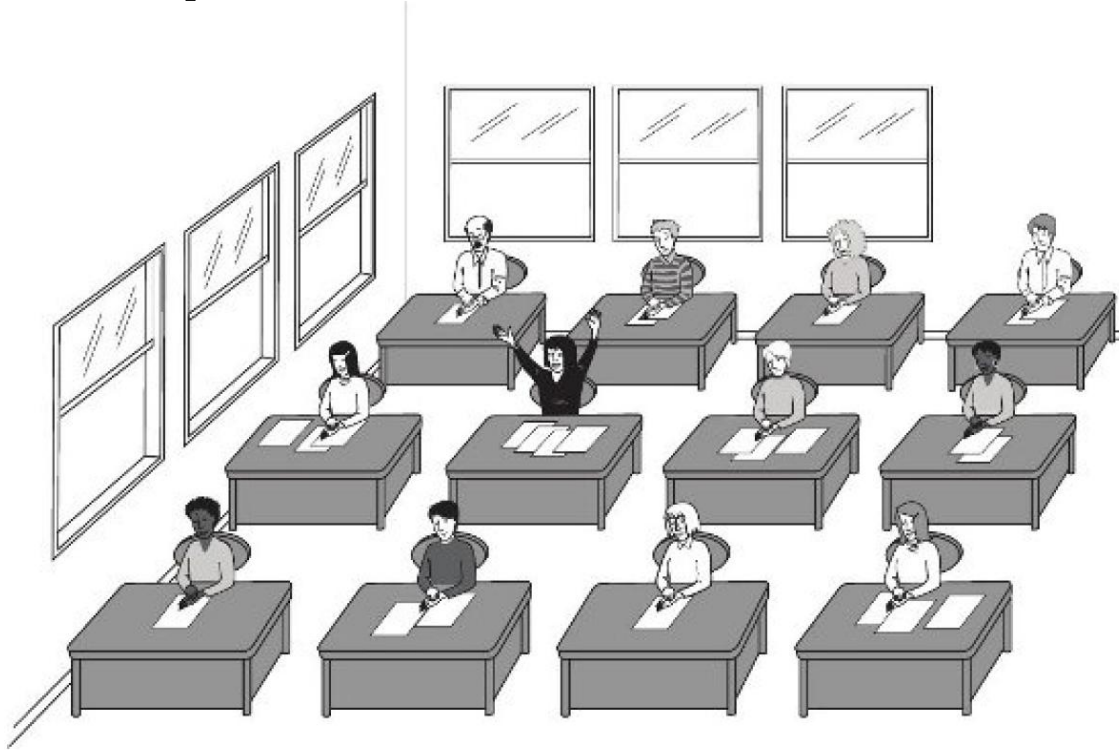
*Taylor series*



# Self Organized Criticality

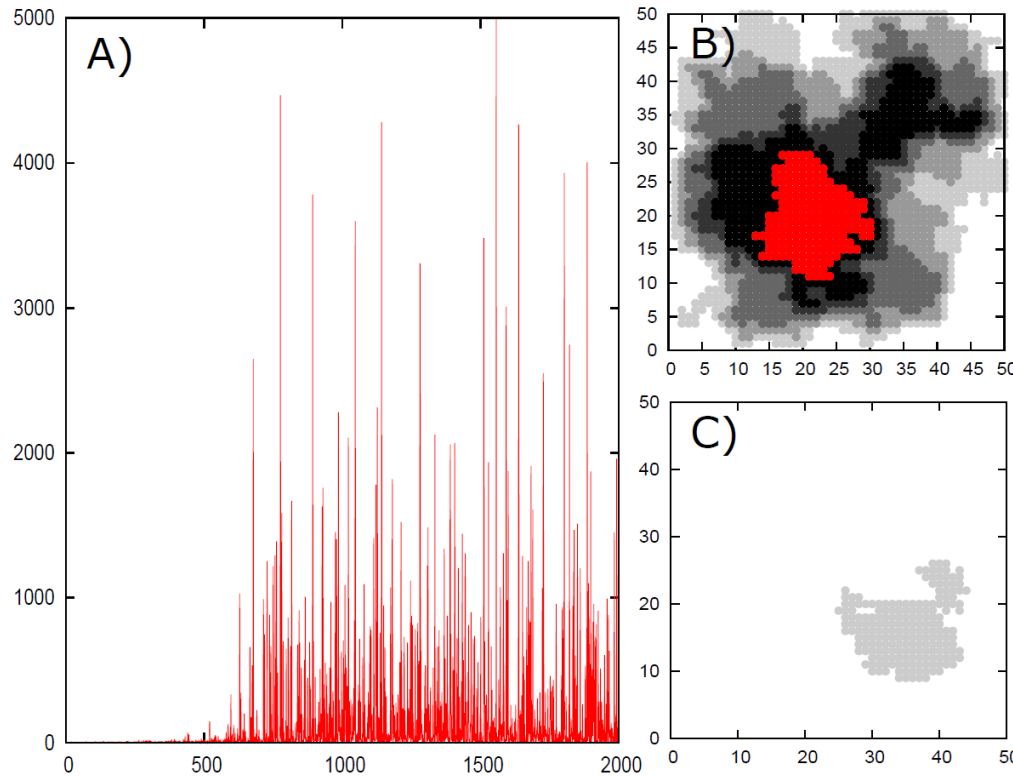
- Tutorial: Why is it desirable to have a near critical chain reaction in a nuclear reactor?

# Sandpile model=bureaucrat model



The canonical version of SOC takes place on a two-dimensional square lattice consisting of  $N = L \times L$  sites. Each site  $i$  can take certain integer values  $h_i$ , where  $h_i = 0, 1, 2, 3, 4, 5, \dots$ . All sites with  $h_i = 4, 5, \dots$  are considered unstable and topple simultaneously. When they topple they distribute one unit to each of their four nearest neighbors. Thus, the sum  $\sum_i h_i$  is conserved when we are away from the boundaries. However, any site that is at the boundary, distributes a unit out to imaginary neighbors outside of the system. These units are lost. When all sites  $i$  have  $h_i < 4$  a new grain is added at a random site, and the above procedure is repeated.

# Simulated Dynamics



Accumulated topplings  
at each site in a  
big avalanche  
(red sites >5 topplings)

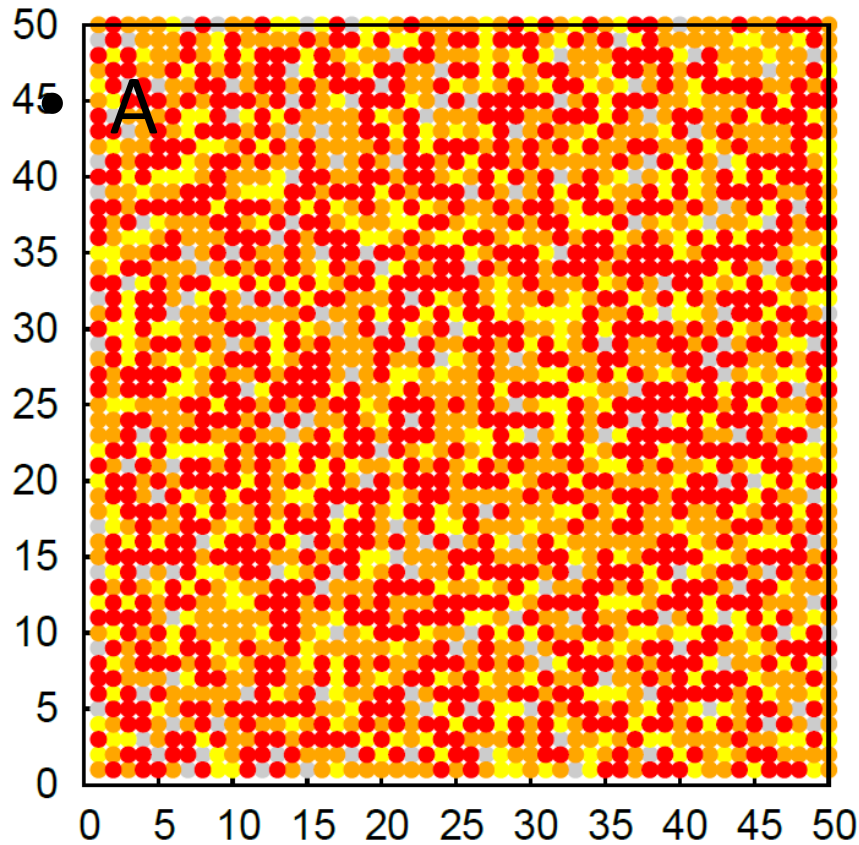
Small avalanche  
that did not even reach  
boundary

Transient build  
up of grain in the system

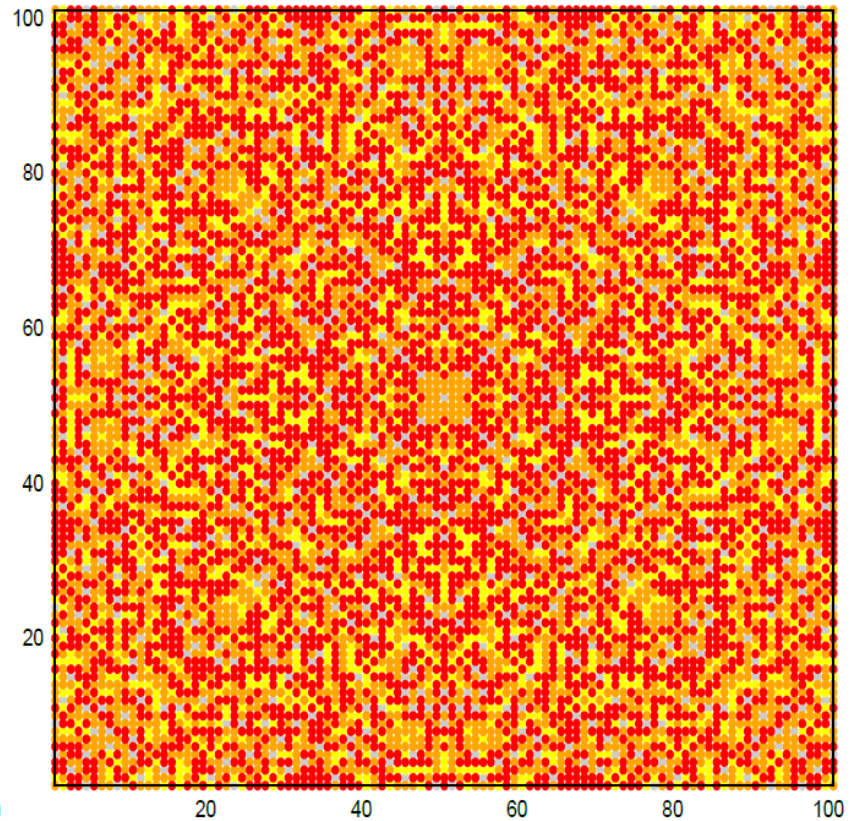
Each line represent  
An avalanche of many  
topplings

.... Self organizing period

# Pictures of critical state:



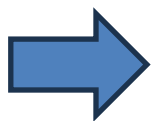
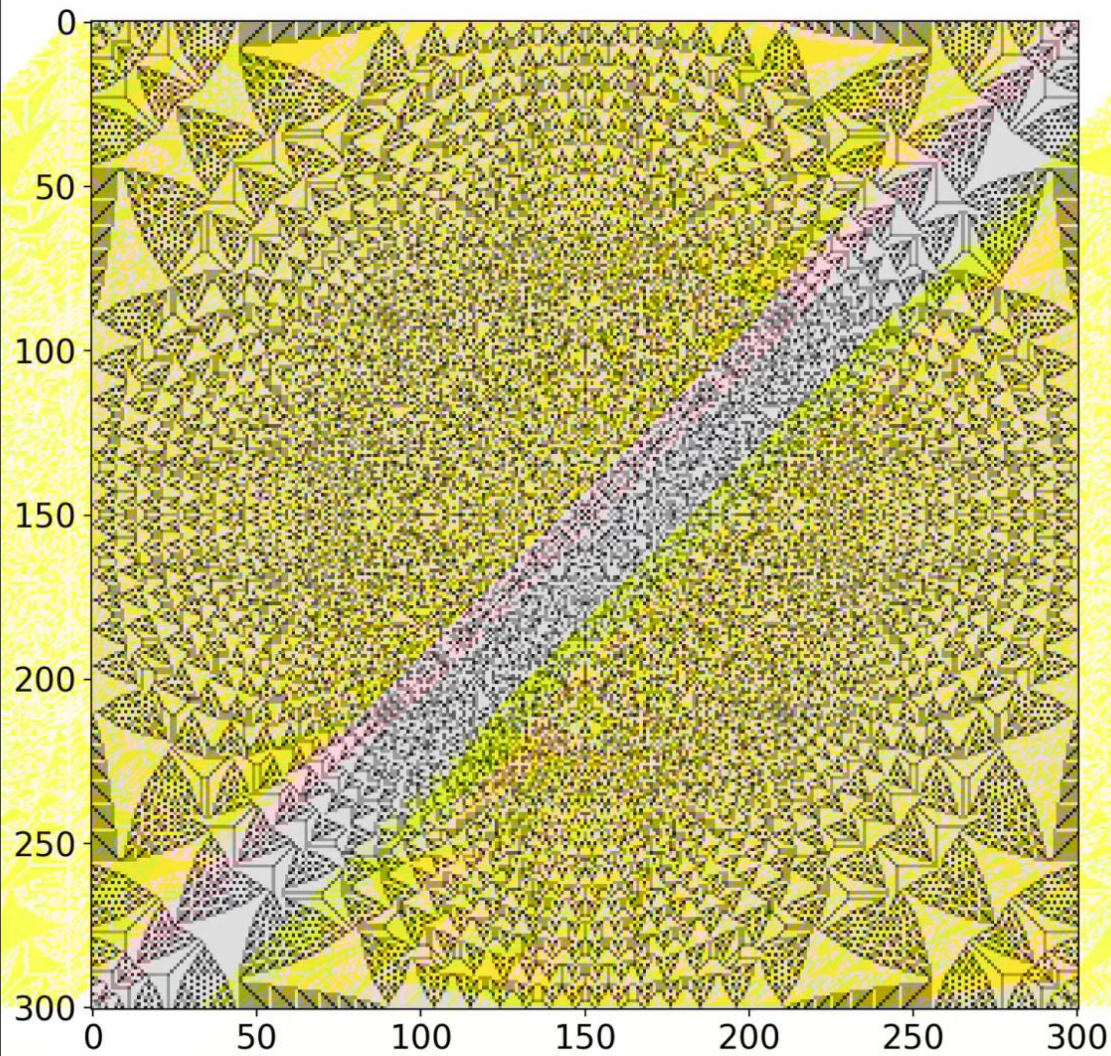
Add randomly



Always add grain at center position



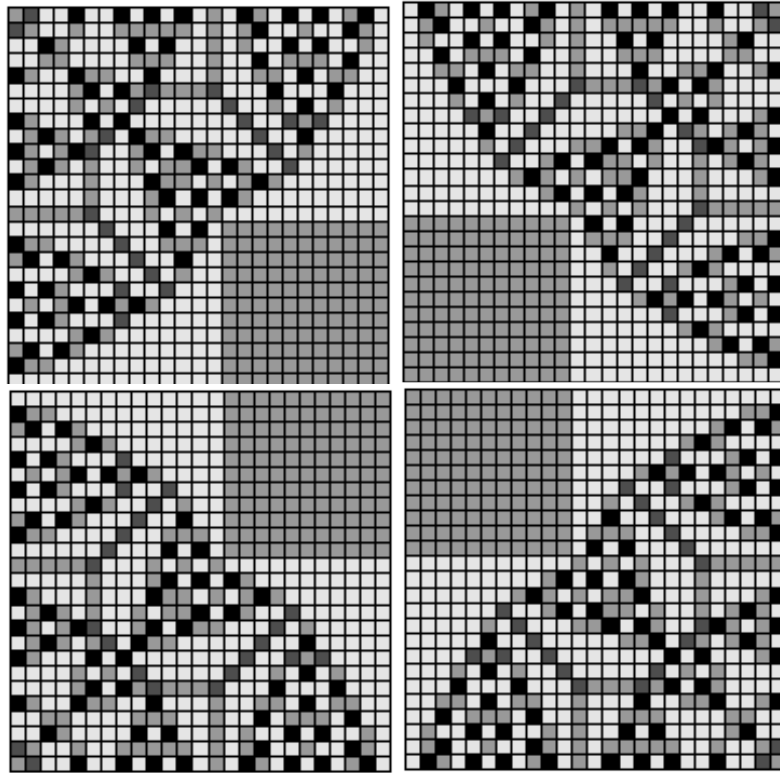
→ Fractals when one always add at the center:



**Long range correlations**

Credits: Marcus Engsig

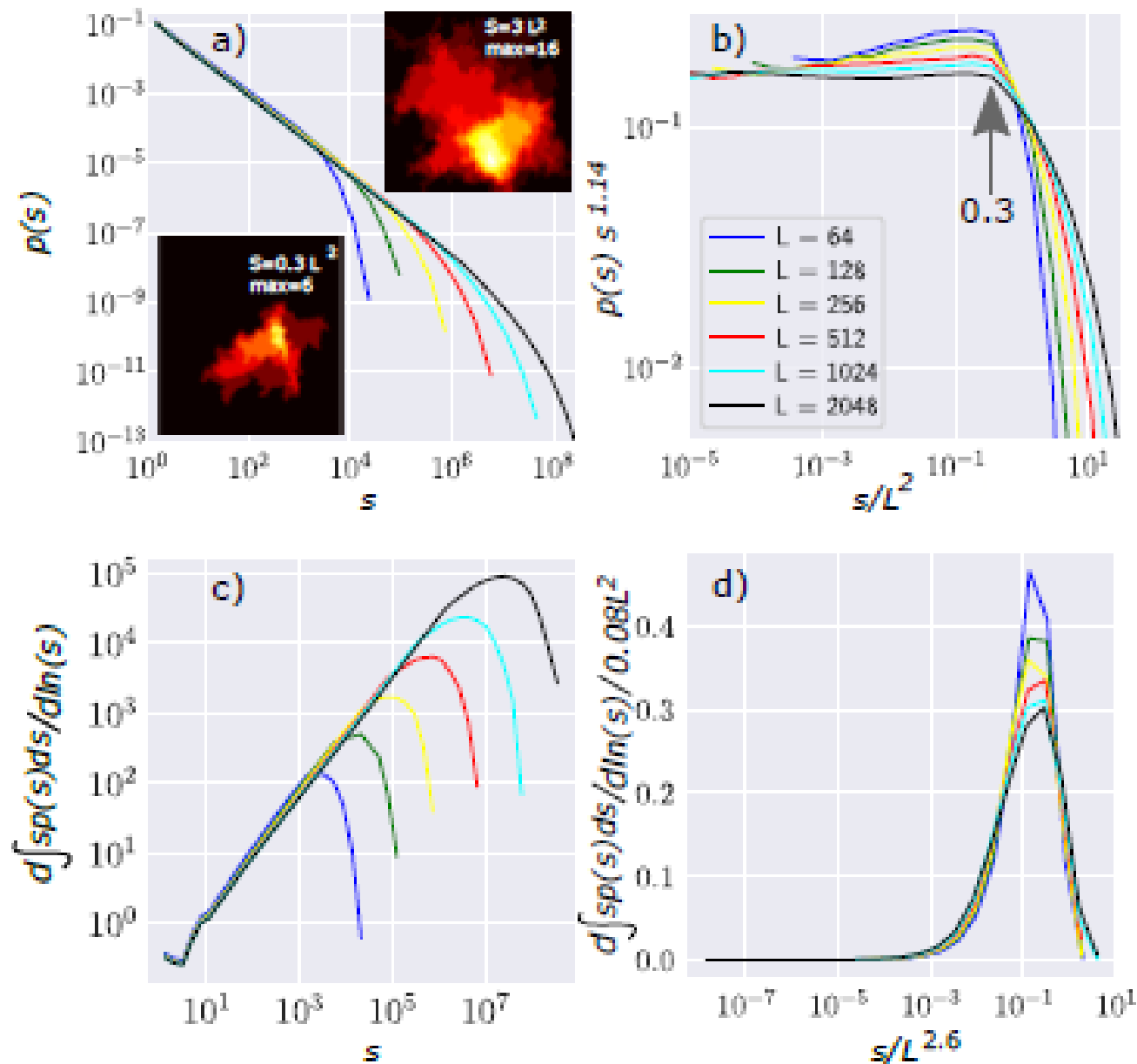
# Identity operator:



# Caviots:

- Only new ``grain” added after full relaxation...  
need separation of timescales
- Need transient of many added grains before  
phenomena are obtained.

# Avalanche size distribution (2d-sandpile)





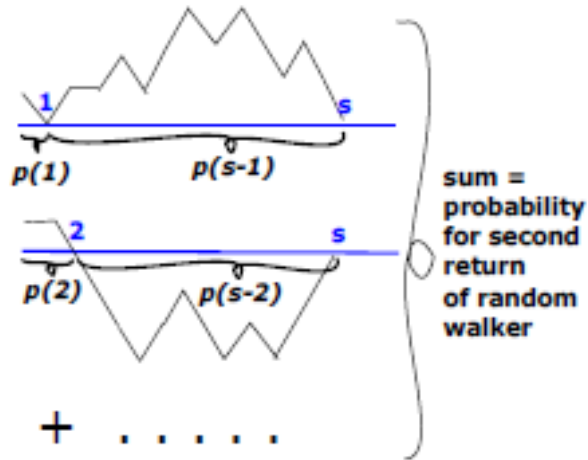


# Recapitulation 23/9-2025:

$$\text{width } \sigma \propto \sqrt{t}. \quad t = \sqrt{t} \int_0^t t' P_{first}(t') dt'$$

$$t = t^{1/2+2-\tau}, \text{ or } P_{first}(t) = t^{-3/2}$$

Folding two subsequent first returns of random walkers



$$p(s) = \sum_{k=1}^{s-1} p(k) \cdot p(s-k)$$

$$p(s) \propto 1/s^{3/2}$$

If you start with filled site:

1:  $p(1)=1/4$

2:  $p(2)=2/2^4=1/8$

3:  $p(3)=5/64$

$$p(3)=(1/4) \cdot (p(2)+p(1) \cdot p(1)+p(2))=5/64$$

4:  $p(4)=7/256$

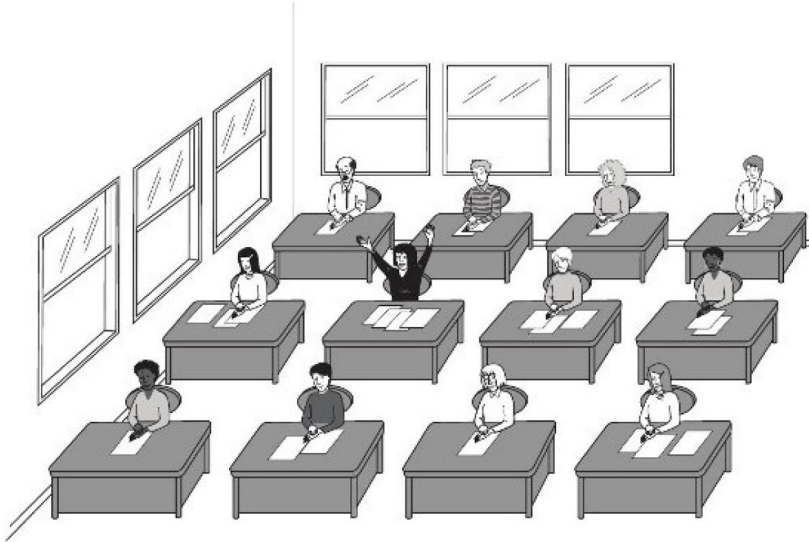
$$p(4)=(1/4) \cdot (p(3)+p(2) \cdot p(1)+p(1) \cdot p(2)+p(3))=7/256$$

5:  $p(5)=(1/4) \cdot (p(4)+p(3) \cdot p(1)+p(2) \cdot p(2)+p(1) \cdot p(3)+p(4))$

$$p(s) = \frac{1}{4} \sum_{k=0}^{s-1} p(s-k-1) p(k)$$

where we define  $p(0)=1$

# Recapitulation 23/9-2025:



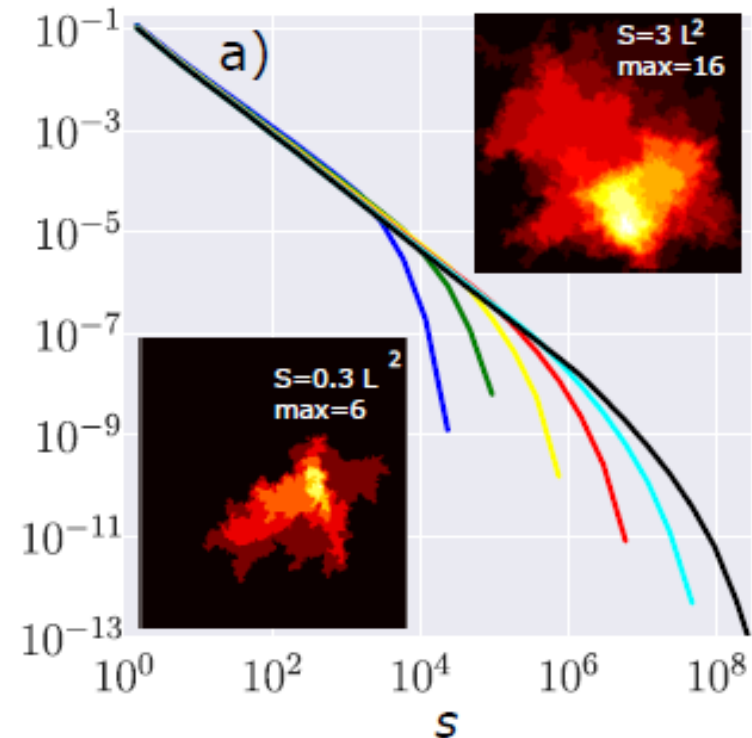
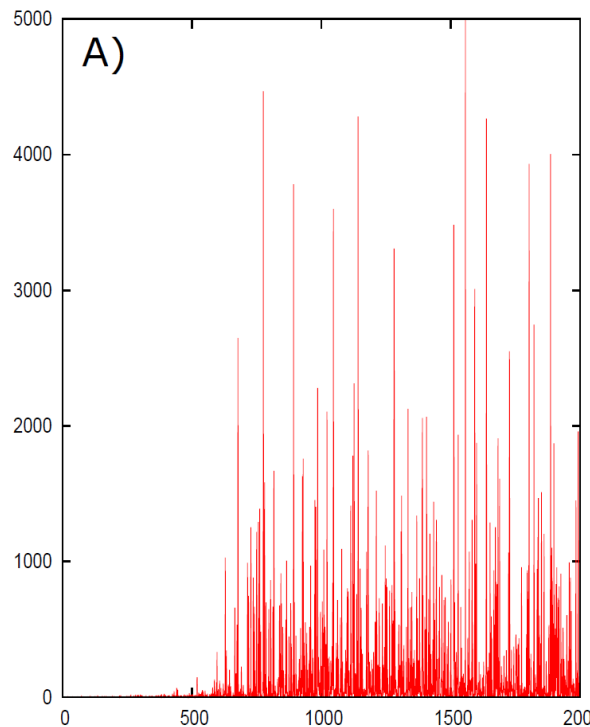
SOC-sandpile needs:

Very slow driving

Transient needed

Transport of stress from input to output  
(windows)

(what happens if one closes windows?)



# Recapitulation 23/9-2024:

SOC – possible in systems with stick-slip dynamics

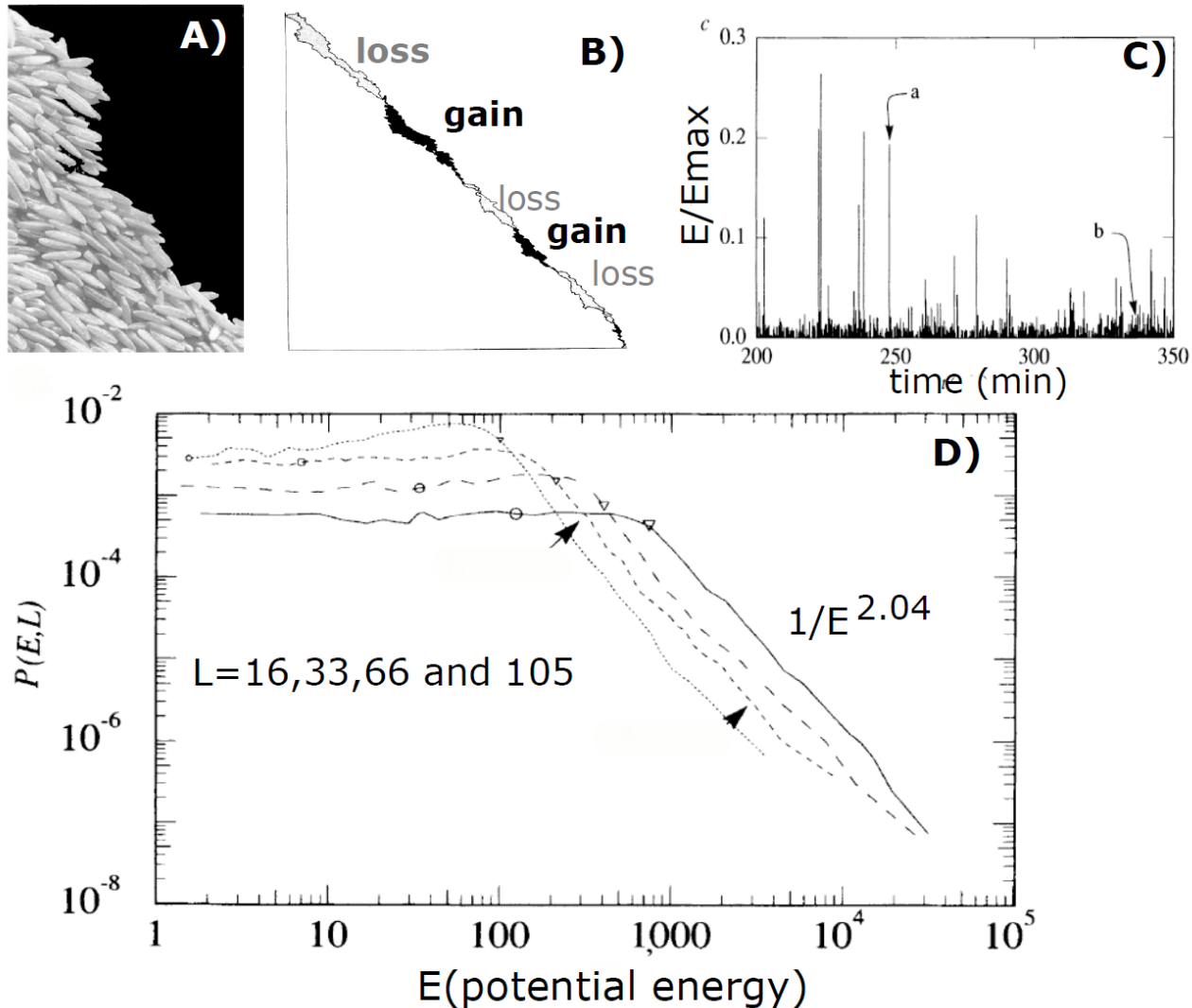
Need separated timescales &  
And need a long time to reach criticality

Nearly always one obtains:

$$P(s) \propto \frac{1}{s^\tau} \quad , \quad \text{with } ; 1 < \tau < 3/2$$

But exponents can go higher...

# Experiment (1-d rice pile)



# Exponent $\tau$ depends on excitation-dissipation geometry

$$P(s) = \frac{1}{s^\tau} \cdot f\left(\frac{s}{L^D}\right) \sim \frac{1}{s^\tau} \cdot \exp\left(-\frac{s}{L^D}\right)$$

Approximation,  
Actually different cut off  
function

Illustrate cutoff at  
 $s=L^D$

Avalanches is activity that allows excitation to move dissipation to boundary:



Following a single sand grain  
as it moves as the current  
site it resides topples.  
Its random walk terminates  
at the boundary, typically  
a distance  $L/2$  away from  
the initial site of addition.

# Number of topplings from random point to edge $\sim 0.08L^2 \sim L^2$

Per avalanche, steady state: One gain **in** gives one grain **out**  $\rightarrow$

$$L^2 = \int_0^\infty s \cdot P(s) ds \sim \int_0^{L^D} s^{1-\tau} ds = L^{D \cdot (2-\tau)}$$

Where we simply replace cutoff function  $f(x)$  with step function at  $x=1$

$$2 = D \cdot (2 - \tau) \Rightarrow \tau = 2 - \frac{2}{D}$$

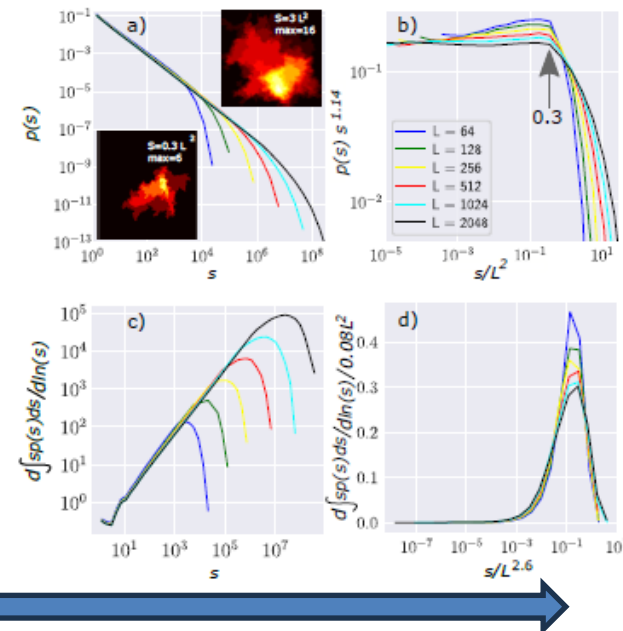
Thus **D=2.6** would be consistent with  **$\tau=1.23$**

$$d \int s p(s) ds / d \ln(s) / 0.08L^2$$

$$\sim \int_{(from\ s\ to\ 2s)} x p(x) dx / \log(2)$$

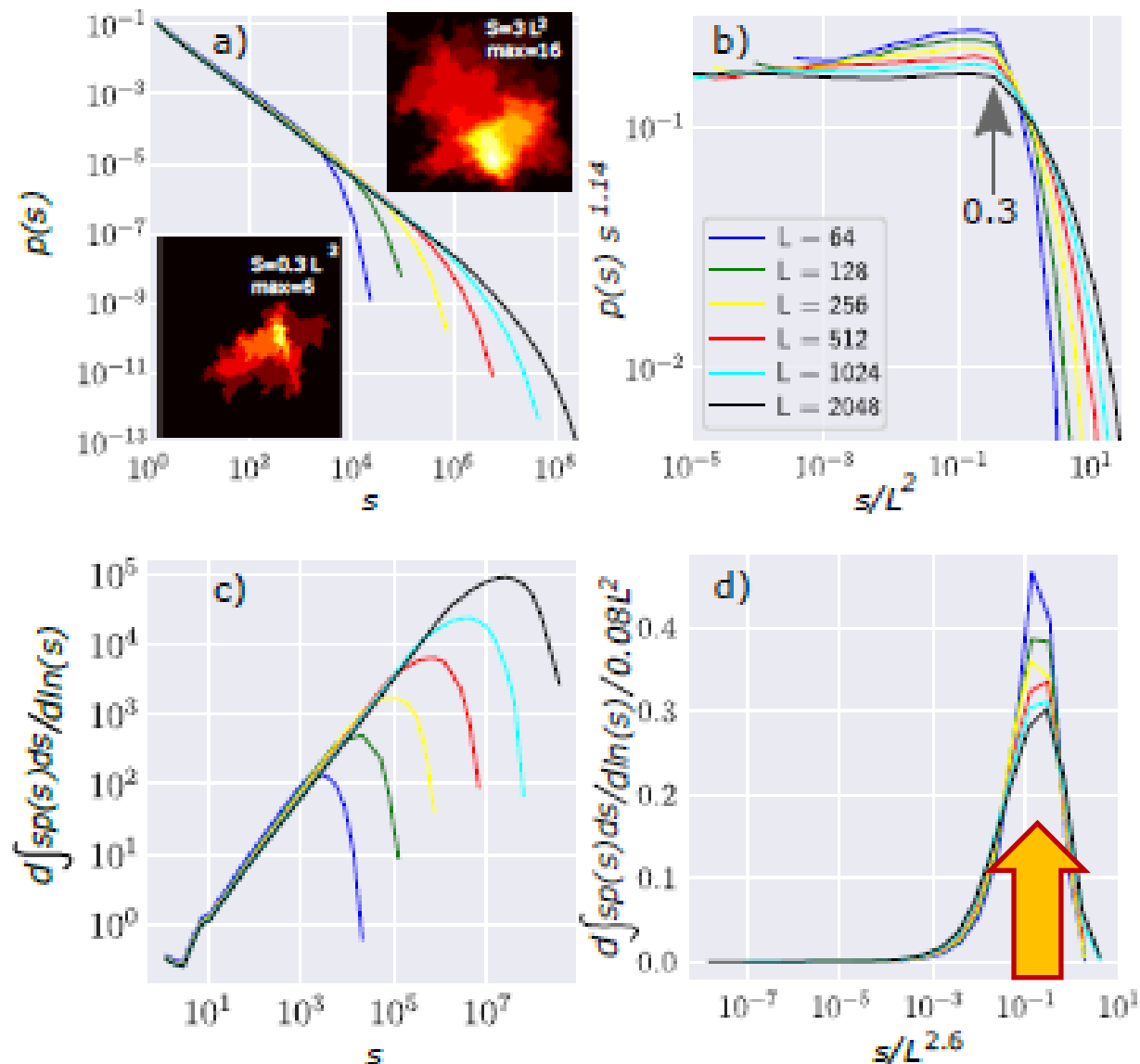
=energy dissipation on scale **s** to **2s**

as function of **s**



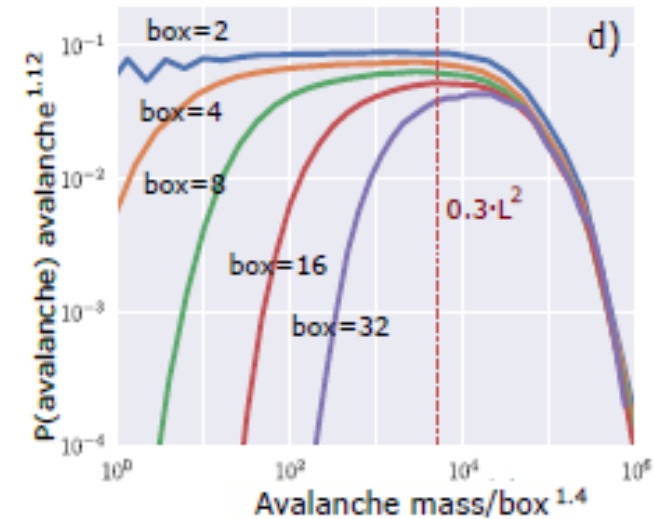
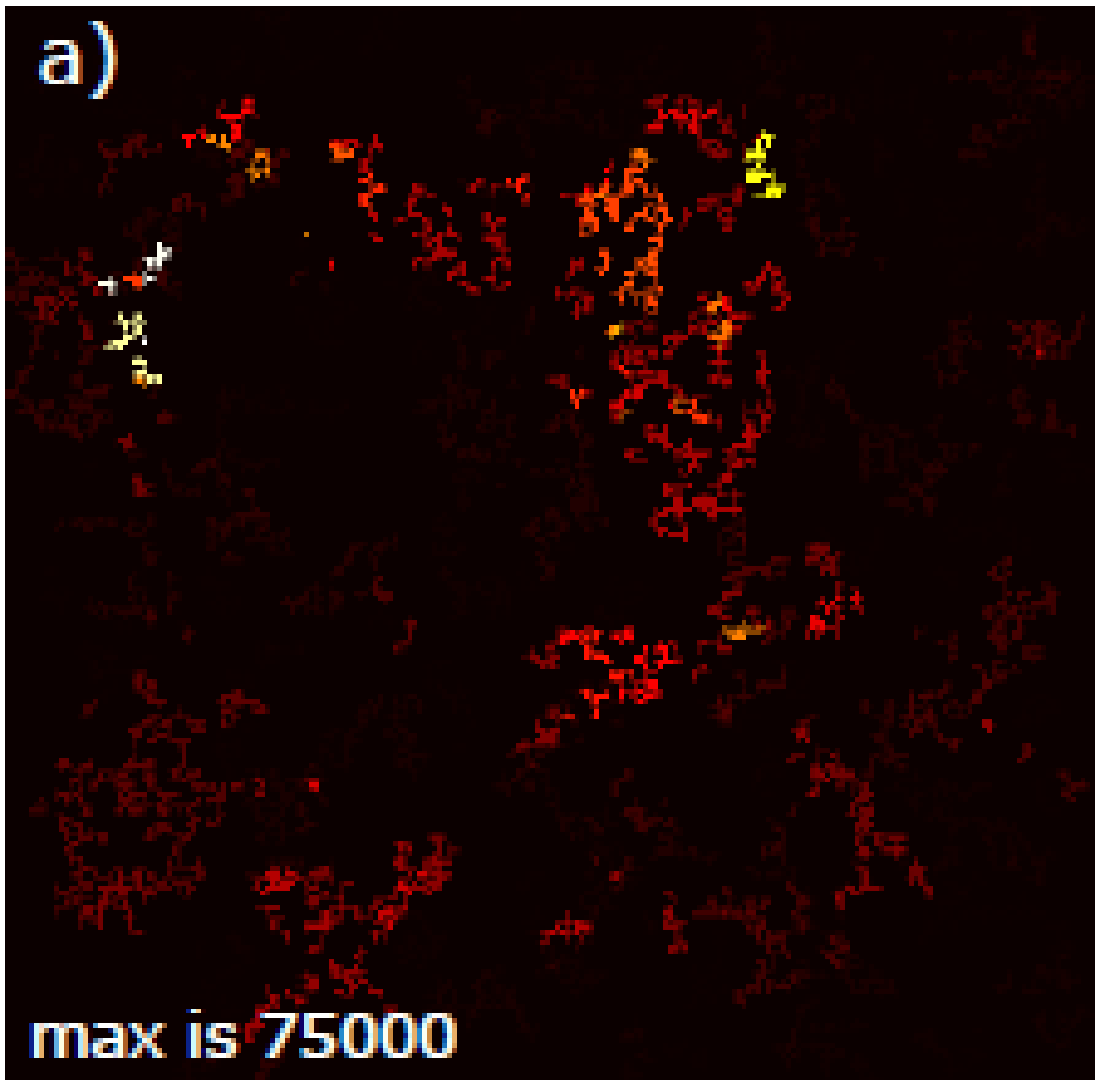
# Avalanches in Abelian sandpile

dissipation is beyond the cutoff scale:





# Highly excitable sites sits on fractal $D_s=1.4$



Take  $L \times L$  start sites, each  
in average giving  $\langle s \rangle = L^2$   
topplings  $\rightarrow$

$$D_s + D = 4$$

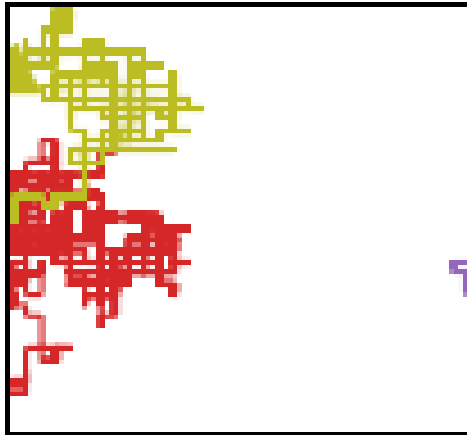
Start  $L^{D_s}$  large avalanches  
Each of size  $L^D$

# If one only adds grain at the edge

They leave faster:

$$\text{average exit time from boundary} = \int^{L^2} \frac{t \cdot dt}{t^{3/2}} \propto L$$

Where upper cutoff is time to reach opposing boundary,  
and the 3/2 exponent is exit from entry boundary



$$L \sim \int_0^\infty s \cdot P(s) ds \sim \int_0^{L^D} s^{1-\tau} ds = L^{D \cdot (2-\tau)}$$
$$\Rightarrow 1 = D \cdot (2 - \tau) \Rightarrow \tau = 2 - \frac{1}{D}.$$

# If one only adds “sand” in one corner:

The probability that it passes one boundary after  $t$  steps is  $\int_t^\infty dt/t^{3/2} = 1/\sqrt{t}$ ,

Notice normalization, the equation also true for time  $t=1$  (it has to leave at some point)

Probability that it does not pass any of 2 boundaries before time  $t$ :  $\left(\int_t^\infty dt/t^{3/2}\right)^2 = 1/t$

First exit of walker starting at corner is  $1/t^2$

$$\text{average exit time from corner} = \int_0^{L^2} \frac{t \cdot dt}{t^2} = \ln(L^2) \propto \ln(L)$$

The avalanches initiated only by excitation at the boundary gives:

$$\begin{aligned} \ln(L) &\sim \int_0^\infty s \cdot P(s) ds \sim \int_0^{L^D} s^{1-\tau} ds = L^{D \cdot (2-\tau)} \\ \Rightarrow 0 &\sim D \cdot (2 - \tau) \Rightarrow \tau \sim 2. \end{aligned}$$

After addition  
of  $2^{18}$  grains  
in same point:

Center driven

$$L^2 \sim \int_0^\infty s \cdot P(s) ds \sim \int_0^{L^D} s^{1-\tau} ds = L^{D \cdot (2-\tau)}$$

$$2 = D \cdot (2 - \tau) \Rightarrow \tau = 2 - \frac{2}{D}$$

Boundary driven

$$L \sim \int_0^\infty s \cdot P(s) ds \sim \int_0^{L^D} s^{1-\tau} ds = L^{D \cdot (2-\tau)}$$

$$\Rightarrow 1 = D \cdot (2 - \tau) \Rightarrow \tau = 2 - \frac{1}{D}.$$

Corner driven

$$\ln(L) \sim \int_0^\infty s \cdot P(s) ds \sim \int_0^{L^D} s^{1-\tau} ds = L^{D \cdot (2-\tau)}$$

$$\Rightarrow 0 \sim D \cdot (2 - \tau) \Rightarrow \tau \sim 2.$$



# Random neighbor sandpile:

- N sites. Each step adds a grain to any site. If a site contains more than 1 grain: It topples and sends 2 grains to other random sites. One of them is eliminated with probability  $1/N$ .
- Critical state where  $\frac{1}{2}$  of sites have 1 grain, and another half have 0 grain (like in the Bethe lattice at percolation).
- $\rightarrow$ self organized critical branching  $\rightarrow$  Avalanche distribution:

$$P(s) \sim \frac{1}{s^{3/2}} \cdot \exp(-s/N)$$

(like when selecting a point of critical percolation on Bethe lattice, and asking what its cluster size was)

# Variant sandpile model: Directed

consider lattice where each site have 2 downstream neighbours (see right)

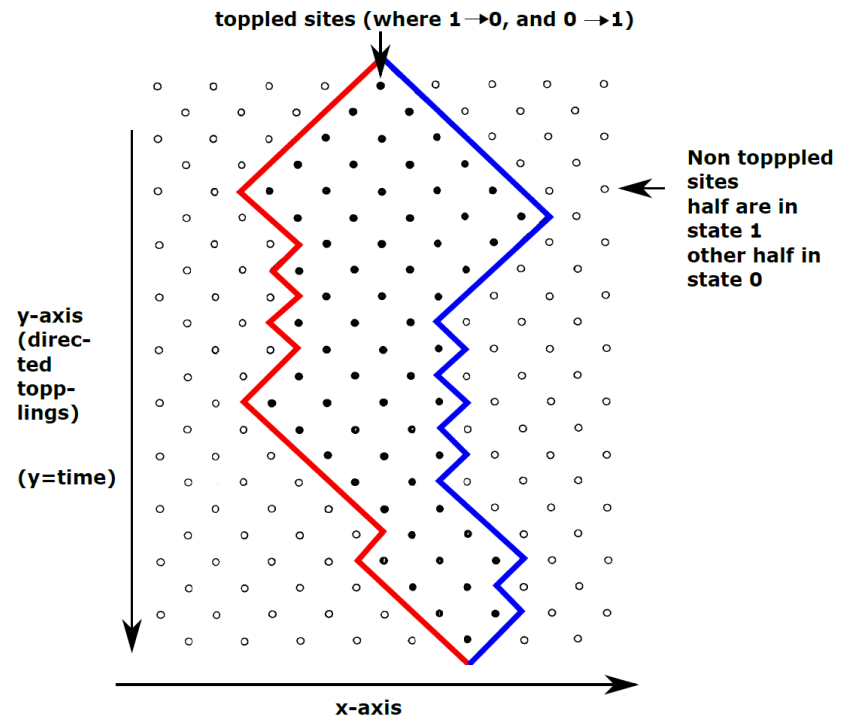
At each avalanche start one add 1 grain at a top Row.

Any site with  $>1$  site topples and give Grain to 2 sites below it.

Gains leave a bottom.

Critical state= 0 and 1 equally distributed randomly at all lattice points.

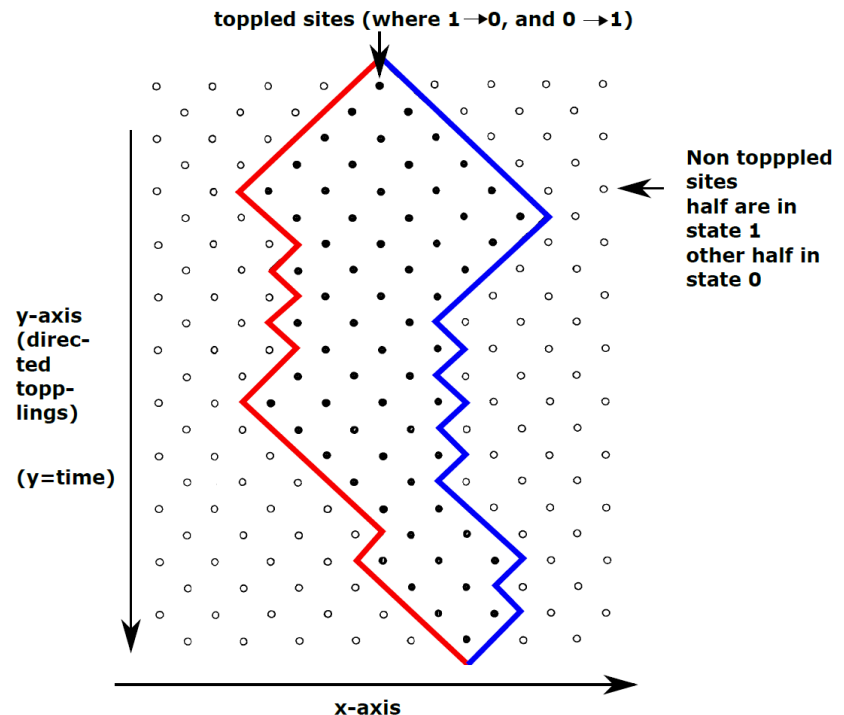
Picture show an avalanche that have to be compact.



# Solving directed sandpile model:

**Compact**, because

A hole would mean that  
It was only receiving one grain,  
Thus only one of the above  
parents could have been  
excited, i.e. it have to be on  
boundary.



0 and 1 equal  $\rightarrow$  boundary makes a random walk

Because a grain falling on a “1-state” boundary site would lead to an expansion of Avalanche.

# Solving Directed sandpile Model:

Random walk boundary  $\rightarrow$  derive exponent

Spatial extension of avalanche makes random walk:

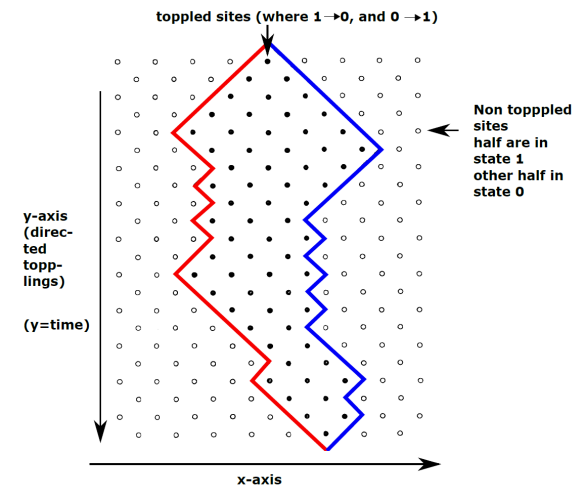
Terminate at time  $t$ ,  $p(t) = 1/t^{1.5}$

Terminate after  $t$ ,  $P(>t) = 1/t^{0.5}$

If size  $t$ , number of topplings  $S = \text{area} = t \times t^{0.5}$

Probability larger than  $S$ ,  $P(>s) = P(>t=s^{2/3}) = 1/s^{1/3}$

Probability for avalanche of size  $s$ :  $p(s) = 1/s^{4/3}$





# Recurrent lesson

**SOC:**

approximately self organize to

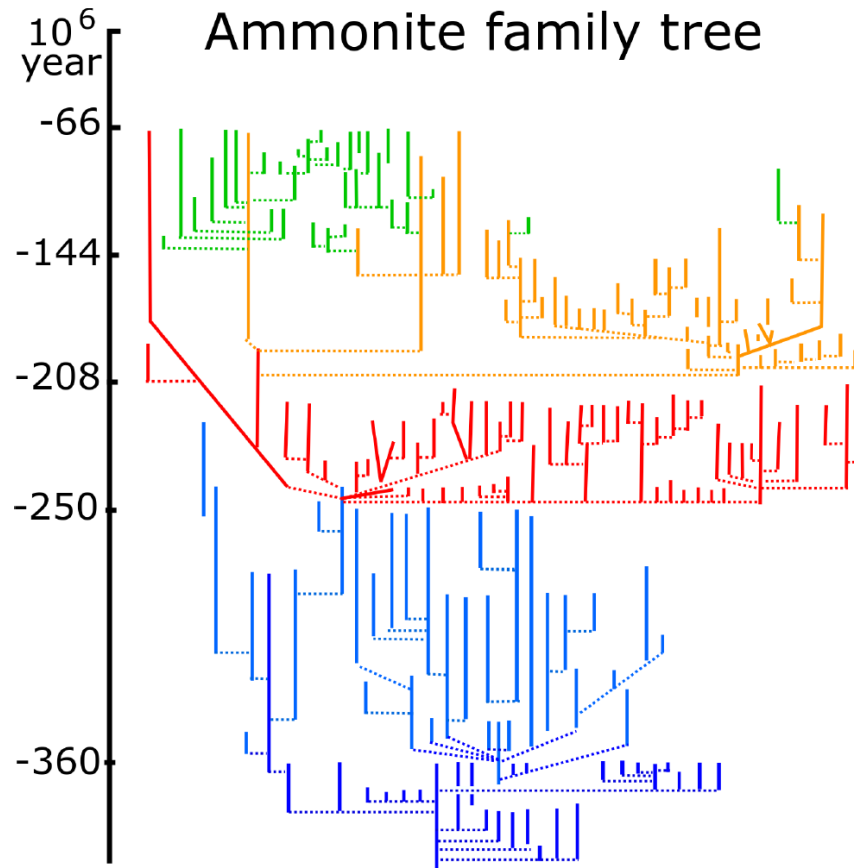
Critical Branching

First returns of random walk

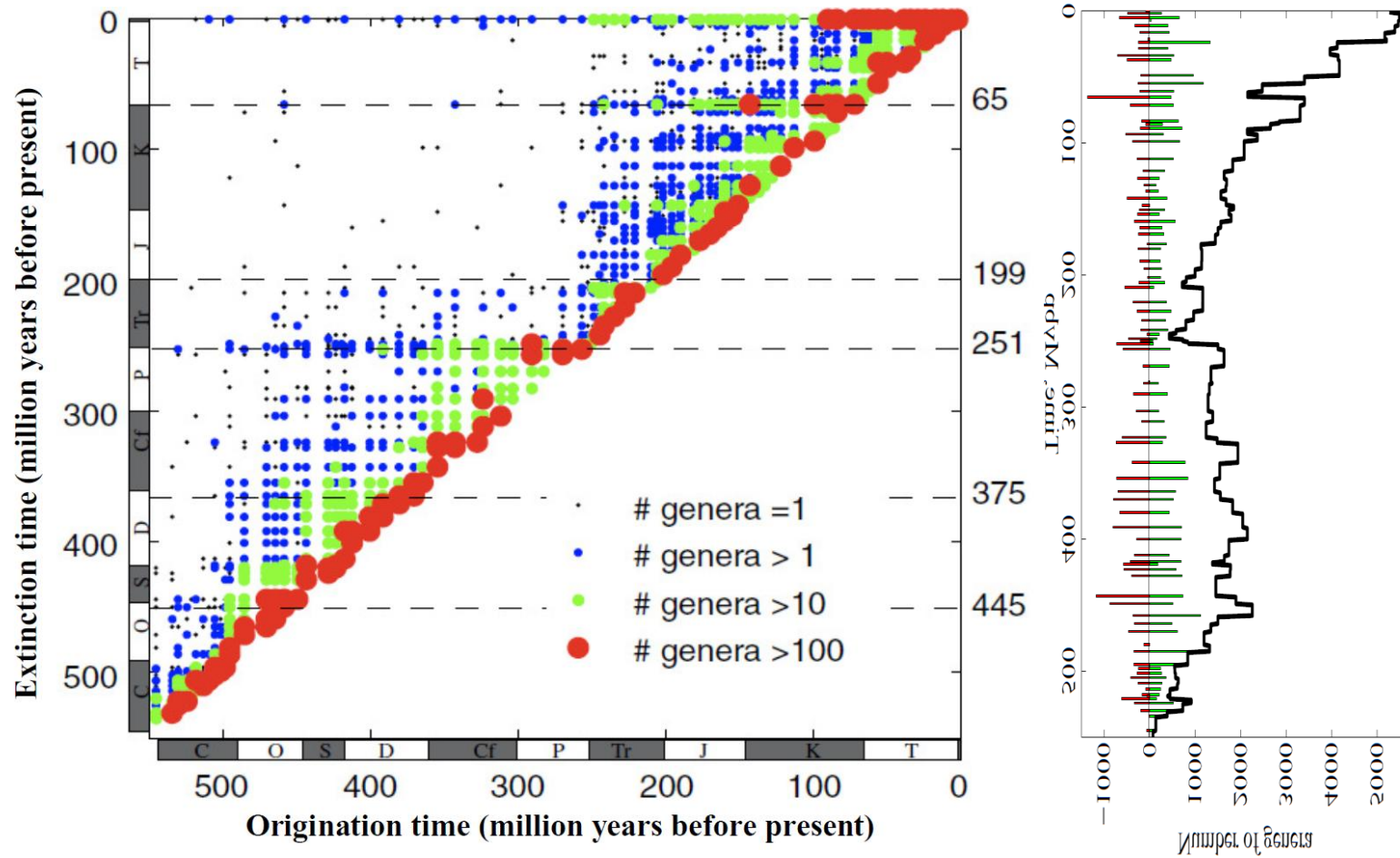
# Self organized criticality & extremal dynamics

**...to do or not to do... the weakest link**

Evolution:  
Stochastic  
...  
With long  
Periods of  
Stasis  
Interrupted  
by  
Huge changes

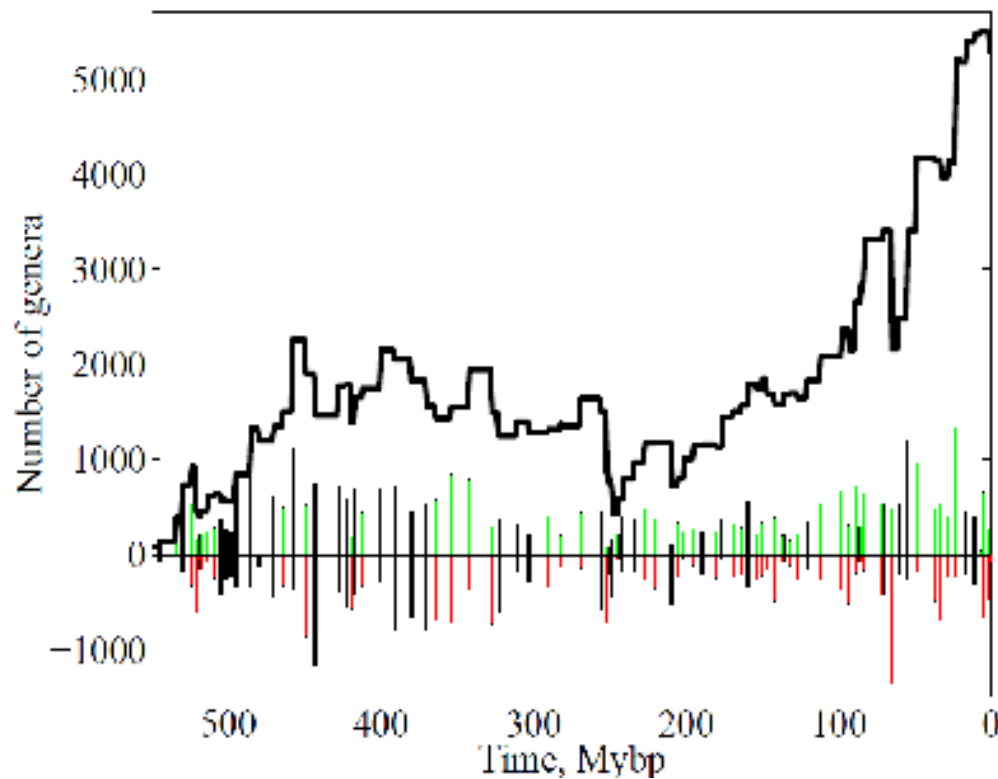


# Extinction-origination matrix

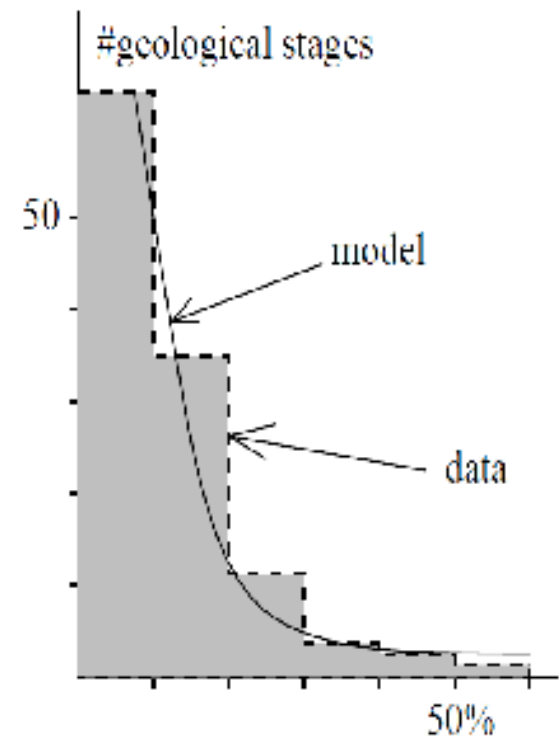


# Origination & extinction last ½ billion year

Diversity and bars for events



Extinction histogram (family level)



# Coarse-grained model:

*Population dynamics*

*Evolution of species*

large  $\xrightarrow{\text{time}}$  scale

*Survival of fittest*

*Barrier dynamics*

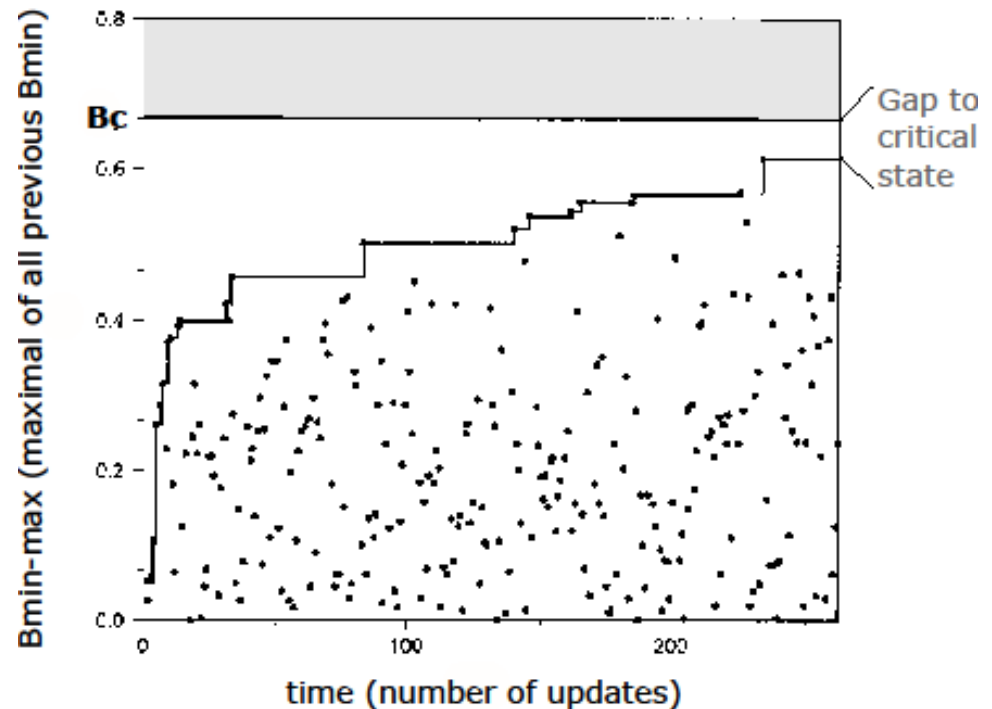
On the scale of a year

On a scale of ~ 10000 years

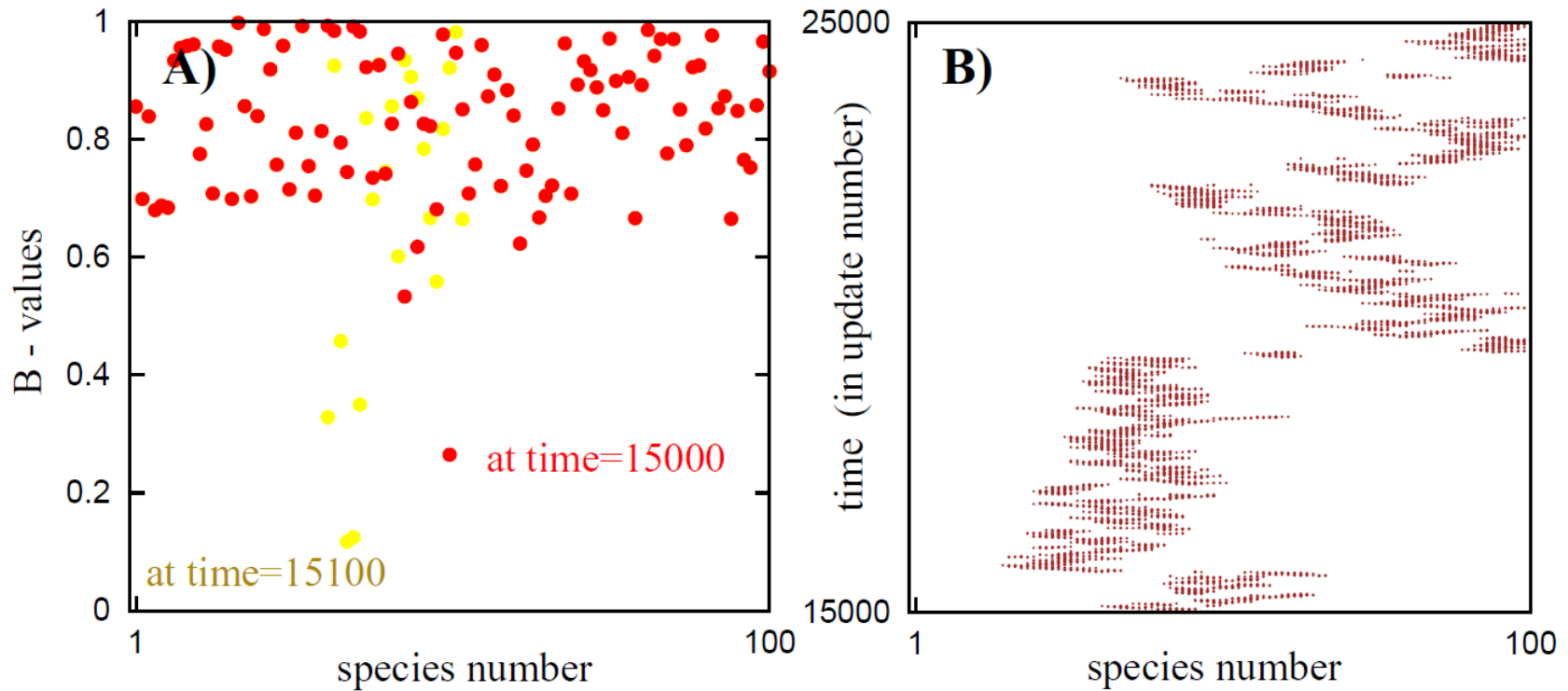
# Model:

At each step, the smallest of the  $\{B_i\}_{i=1,N}$  is identified. For this as well as its nearest neighbors one replaces their  $B_i$ 's by new random numbers in  $[0, 1]$ .

Long way to criticality

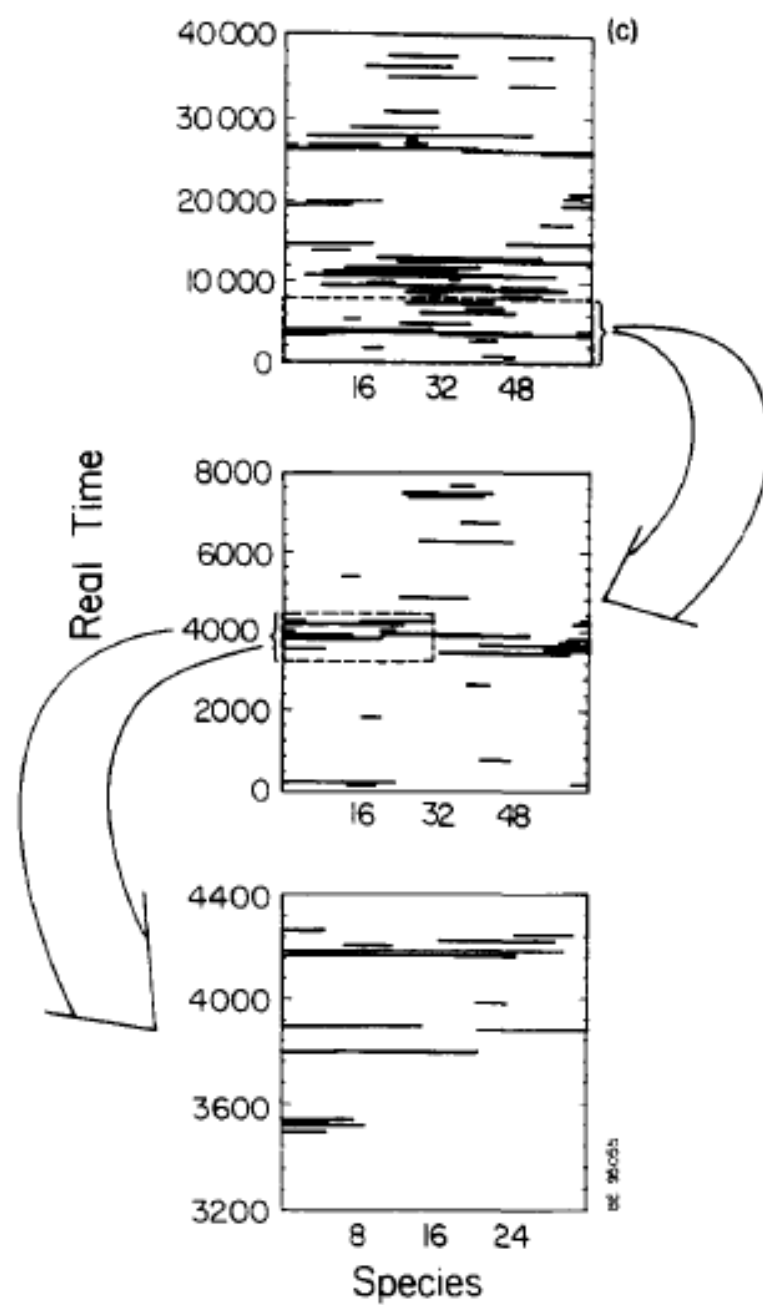
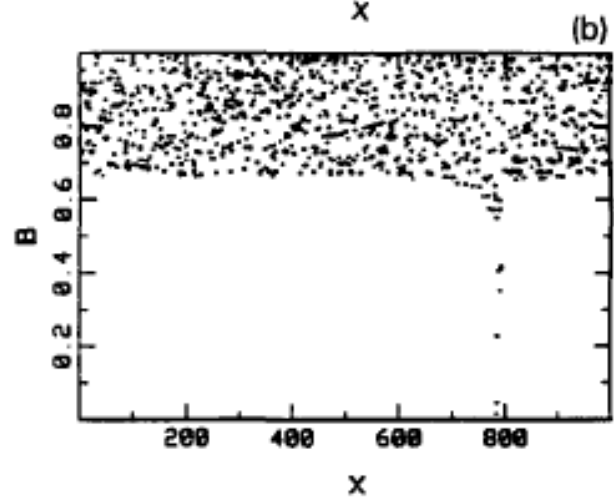
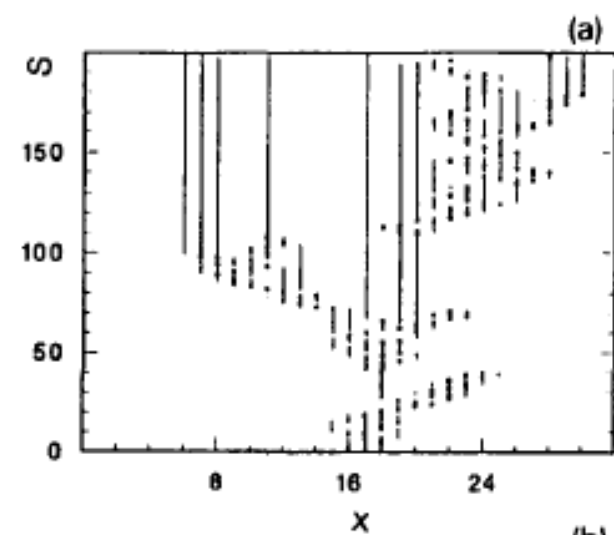


# In Critical state



Left: Yellow dots= latest 20 updates

Right: 10000 updates, position of  $B_{min}$

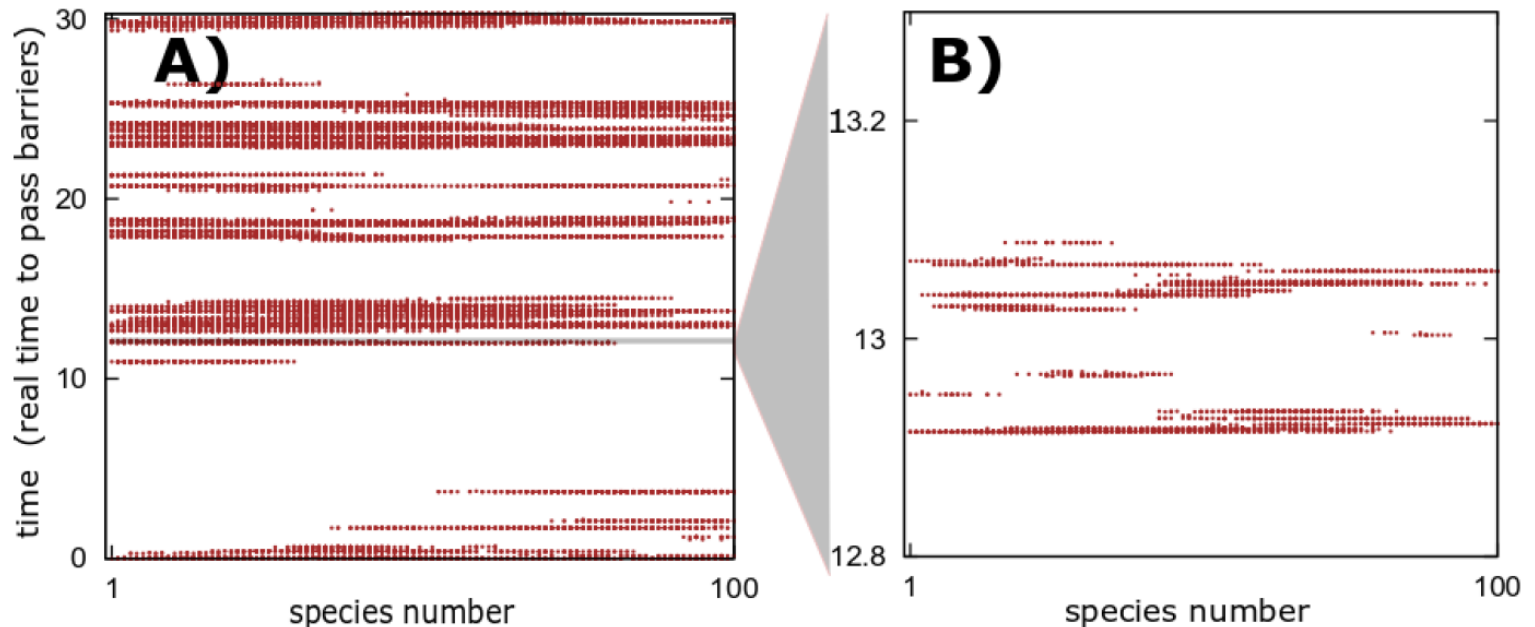




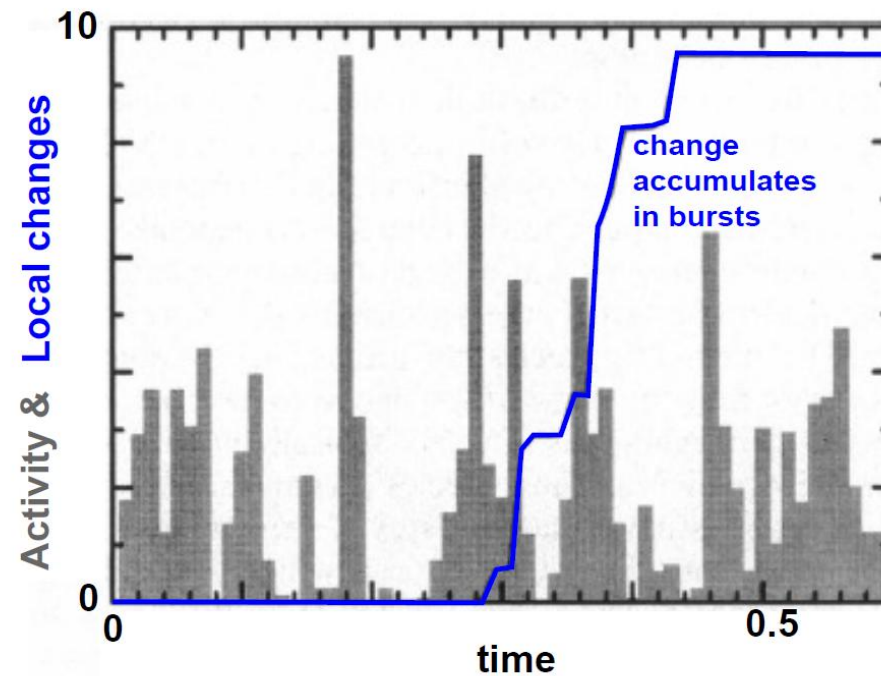
# Why $B_{\min} \rightarrow$

## Rare escapes & separated timescales

- At each time step of size  $dt$ :  
Select each of the  $\{B_i\}_{i=1,N}$  with probability  $\propto e^{-B_i/\mu} dt$ . This selection defines a list of active sites. For members in this list, replace them as well as their nearest neighbors by new random numbers in  $[0, 1]$ .



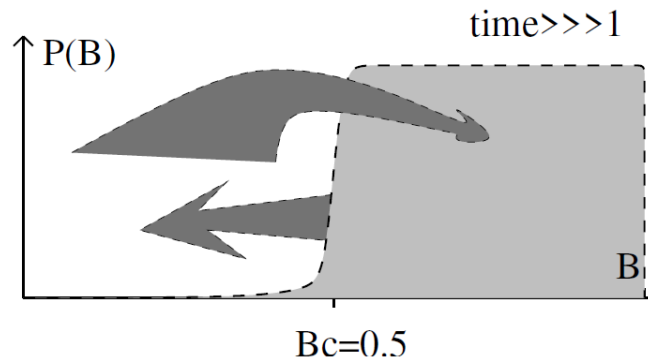
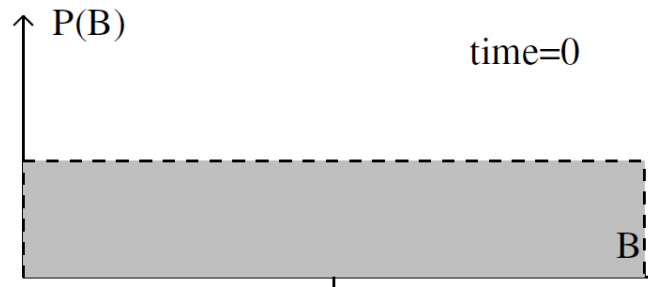
# Accumulated number of updates on one site:



→ **Change in one site** occurs in bursts: Punctuated equilibrium

# Random neighbor version

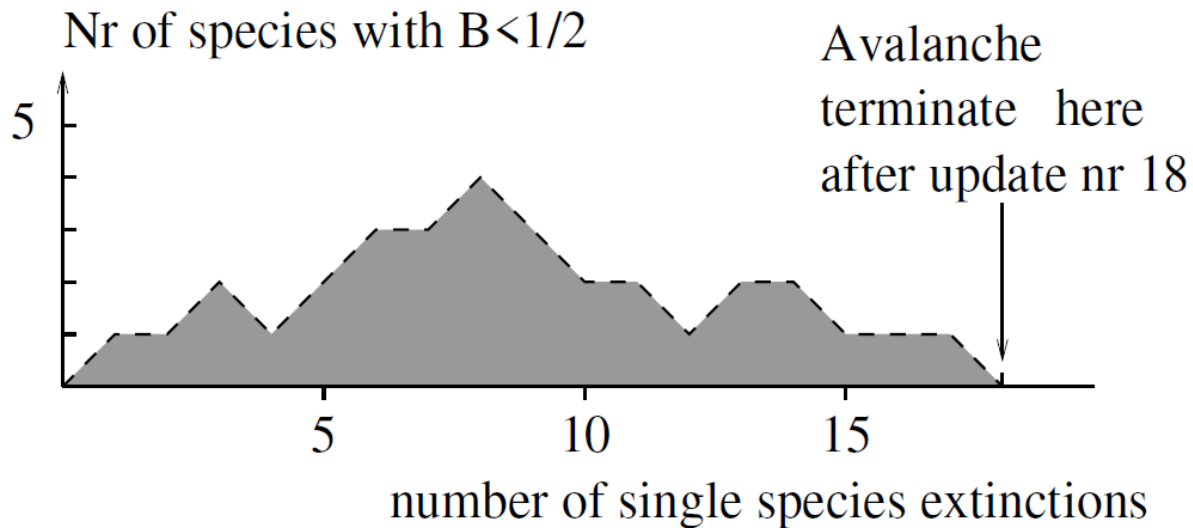
- *At each update, change the minimal  $B$  and one random other  $B$  to new numbers in  $[0,1]$*



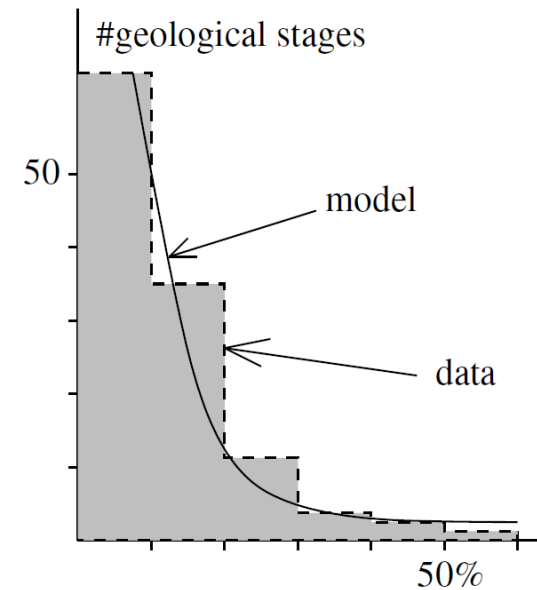
Steady state=Critical state,  
When gain and loss below  $B_c$   
balances:

$$-1 + 2 B_c = 0 \Rightarrow B_c = \frac{1}{2}$$

# Random neighbor → First return statistics on the number of B's below $B_c$



$$P(s) \propto s^{-3/2}$$



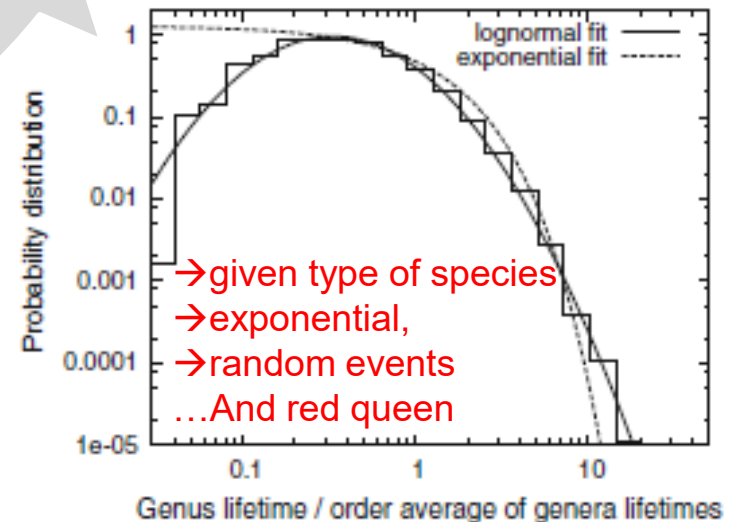
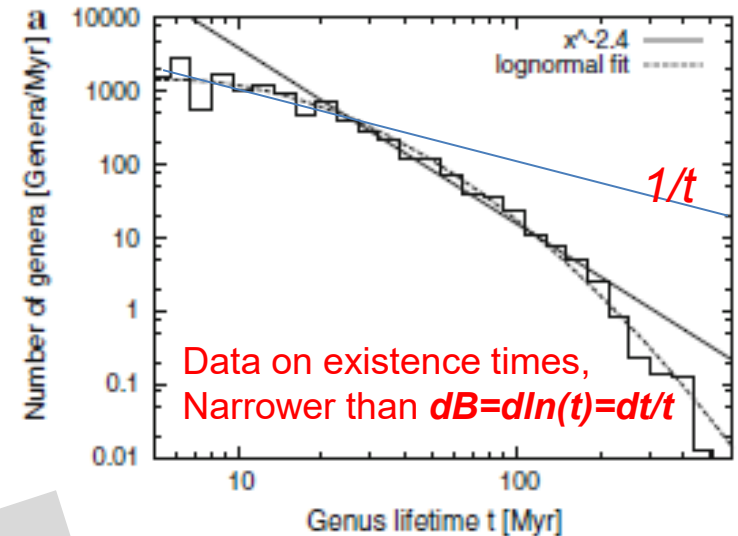
# Problems in mapping model to macroevolution:

$$\Delta t \propto \frac{1}{\sum_i e^{-B_i/T}} \sim e^{B_{\min}/T} \quad \text{for } B_c - B_{\min} \gg T$$

$$P_0(s, \epsilon) \propto s^{-\tau} F\left(\frac{s}{\epsilon^{-1/\sigma}}\right) \quad \epsilon = B_c - B_t$$

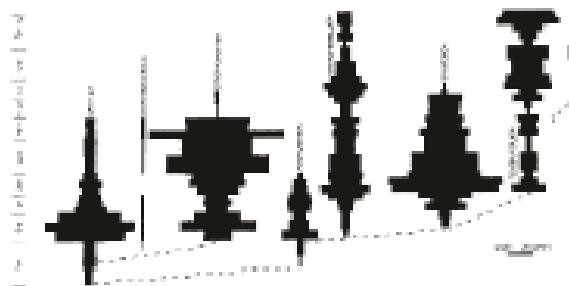
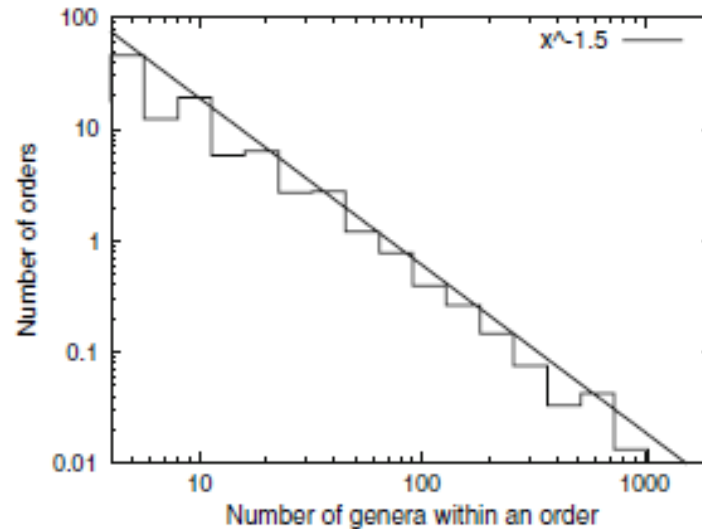
$$T \sim \epsilon$$

$$P_T(s, 0) \propto s^{-\tau} F\left(\frac{s}{T^{-1/\sigma}}\right)$$



# Other power laws in ecology:

(not addressed in SOC picture)



(from Gould et al, 1977)

# Recurrent lesson

*SOC: have Long transients and demand separated timescales*

